IIMB

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THE PM PROCESS

A guide on how to prepare for the Product Management companies that come to IIMB for summer placement



SIGMA

Technology & Product Management club

ANKUR GUPTA

Preface

Hello dear reader. Welcome to this modest attempt at passing on some knowledge, or Gyan as you might call it. The purpose of writing this book is to give you insights into acing the 'Product Management' interview for companies that visit IIMB during the summer-placement internship process.

Some of you might be wondering, '*What does Product Management mean?*' That's ok. Many of us didn't know either what Product Management (called PM from now on) meant and how to go about preparing for the role. We learnt along the way, a lot from material available on the internet and a lot from seniors in our time.

'You mentioned material. If there are resources out there already, why do we need this book?' Good question. The simple answer to that is, while we were preparing for the PM role, we got a lot of unorganized, and sometimes, confusing advice from a lot of sources. Some sources asked us to do something while other sources asked us to do something else. We felt that there was a need for some standardization of the knowledge that was being passed on. Thus, this book is an attempt at standardizing some of the advice and knowledge that we received throughout our own prep and convey it in a way that makes prepping for the PM role more structured.

Keeping that in mind, the chapters ahead are going to be divided in the following format for easy reading:

1) IIMB Placement Process - A summary of the IIMB summer-internship placement process
2) What is Product Management
3) How to begin preparation - An overview of the prep resources that you might be needing while preparing for the role
4) Resume-based Questions
5) HR Questions
6) Guesstimates
7) Product Design/Modification Questions
8) Tech Questions
9) Technical Cases
10) Coding/Algorithm Questions
11) Company-Specific Products
12) Conclusion
13) Appendix 1 - GD Day Cases - Examples of GD-Day cases
14) Appendix 2 - Interviews - Example of actual interviews
15) Appendix 3 - Experiences - Day-to-day life of a PM intern in several companies

Before we begin the serious and fun stuff, a few disclaimers:

Disclaimers:

1) There is no guarantee of you getting that PM internship after reading this book! The book is a resource that is meant to show you how to get there, but you will have to practise the things written here and be prepared for the questions and cases and might come on the GD/Interview day.

2) Since this book is written in 2019, all the advice/opinion/suggestions in the book are valid until May 2019. By some random luck if this book is being read in 2029 and there are companies that are looking to manage flying cars, nuclear fusion reactors or other unheard of products, then they might be looking for different skills in a PM intern, in which case the content in this book might become outdated (unless some kind soul specifically ensures that the content in the book constantly stays updated)

3) And finally, this is an opinion-based work and not meant to be factual in nature. All the opinions mentioned here are either derived from personal experience or from knowledge gathered from seniors. Please don't treat the advice in this guide like it is law. Treat it as a guide that gives you some clarity on how to approach things.

And so with that, let's begin this journey of preparing for a PM role. And while we do that, don't forget to,

Work Hard!

Play Hard!

Stay Motivated!

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Chapter 1

IIMB Placement Process

Welcome, dear reader, to the IIMB summer-internship placement process. This experience might be unlike any other placement experiences you have experienced so far, not only in terms of balancing preparation and academics together, but also socially, and (for some) mentally.

So in order for things to not get overwhelming, let's take things a bit slow and begin right at the top. This chapter is going to give an outline of the IIMB summer-internship placement process. Note that this is a concise summary of the process and misses out on many details.

Placement Process



Resume Preperation

Get the proper resume format and prepare initial resumes. Jot down all points

1

2



Resume Verification

Verify, with proof, all the points in the resume with the PR in the loop

3



Online Application

With the final resume, apply to companies on the placement portal

4



Attending PPTs

Attend company PPTs as and when they are scheduled. Get to know more about the company and the offered role

5



Shortlists

Some time after online applications are done, individual shortlists for companies come out

6



Quizzes & Tasks

Certain companies require you to take online quizzes or complete tasks to proceed to the next round of application

7



GD/Interview Prep

While the application process is going on, prepare for GDs and final interviews

Resume Preparation starts well into the first term itself, with the PRs providing you with the proper format and assigning resume mentors who help you out during the resume building process. After making you do multiple iterations of resume-building, and once the points on your resume are almost finalized, the PRs will ask for proof of each point on the resume.

With the final verified resume, you can then apply to internship companies via the online portal. Each company that comes on campus has different pre-requisites that you might have to consider before applying to those companies. Some might have a work-experience bar, while others might have a background preference (for example, candidates who have completed CFA).

While the online application process goes on, several companies hold Pre-placement Talks (PPTs) that you can attend and get a better idea of the company that is giving the talk and the roles that it is offering.

During this phase of applying online and attending PPTs, some companies already start releasing resume-based shortlists of preferred candidates. Most companies though (and most PM companies) release their shortlists maybe a few days before the actual placement dates. As shortlists come out, some companies might have extra rounds of tasks/activities (such as making a wireframe for a mobile application) that shortlisted candidates will have to complete in order to be eligible for future interview rounds.

Apart from all this prep, you will also have to specifically prepare for the GDs and actual interviews. This is the most important part of the process because no company takes a candidate solely based on their resume. All companies will test your communication and other skills on the GD/Interview day itself, no matter how polished your resume is.

A bit about the roles

With so many companies coming to recruit and so many roles being offered, it helps in dividing the roles into segments so that preparation doesn't get overwhelming. So let's divide the roles into 6 major sectors: 1) Product Management 2) General Management 3) Sales/Marketing 4) Consulting 5) Finance 6) Other roles such as analytics, operations etc. Most companies offer roles that fit neatly into one of the categories above. For example, Microsoft offers PM & Marketing roles. Amazon offers PM, Program Manager, and Operations roles.

By fitting the roles into neat categories, preparation for the roles becomes easier, because now you don't have to prepare for companies, you have to prepare for sectors. During the beginning of prep-season, you will hear a lot of this type of conversation:

A: Which sector are you preparing for?
B: I'm thinking of preparing for Product Management
A: Not finance or consulting?
B: No. I might do some prep, but my primary focus is on preparing for the PM role

Starting preparation in an organized manner begins with identifying the sector you want to prepare for and utilizing resources specific to that sector. About that time, you will notice that prep groups also begin forming. These groups are composed of people with similar sector-interests.

A piece of advice here is that, prepare hard only for one role. The rationale behind this advice is that since preparation for any role takes a lot of time and effort, preparing for more than one role will dilute your effort that you put into each role. For example, preparing for the PM and finance roles together are very hard to do because they require different knowledge sets and diametrically opposite skills.

Chapter 2

What is Product Management

With the big picture out of the way, let's take a step down the ladder and begin focusing on Product Management. For many of you, you might be wondering '*What do the words 'Product Management' mean?*' To this question, there is no well-defined answer. If you try to find the definition online, you will notice that many sources say many things. So let's not rely on online definitions and let's try to figure out what a PM is by the PM way.

The PM way? It means breaking down a given problem into little solvable components, signifying what a PM ideally does on a day-to-day basis. Beginning with our problem, the words 'Product Management' can be broken down into two words: 1) Product 2) Management.

What is a Product? A good or service that a company sells. What is management? What you and I hope to do in a few years with a degree from IIMB in hand. Combining these two definitions, a PM can be defined as being kind of a manager of a product. Some books even call it as being a 'CEO' of the product.

Now being a manager, you will be responsible for the product, which means you will be responsible for:

1) Designing the features of the product
2) Designing and building a product prototype
3) Testing the prototype
4) Making the required changes to the product prototype based on user feedback and usability
5) Designing and developing the final product
6) Maintaining the product

In summary, you will be managing the entire product life-cycle all the way from the design stage to the maintenance stage.

Imagine thinking through something from the start to the finish. The entire process involves a lot of **decision-making** and **planning** and answering hundreds of questions while making those decisions, with some of them being along the lines of:

1) What features should I incorporate in the product?
2) How should I build the prototype?
3) What metrics should I be observing while carrying out prototype usability tests?

Some questions will be open-ended, while others will be specific and close-ended. And with so many questions to answer and decisions to make, you will have to have a **highly-structured**

mindset and **step-by-step thinking** process in order to come up with some logic to your decisions.

As anyone who has built a product previously knows, during the product-building exercise, you will have to **talk to customers** to find out their pain points. Since you have to talk to customers, you will need **communication** skills. During the talking phase, you will have to use your **understanding of the technology** to try to solve the customer's pain. Sometimes the customers might not even know they need the product, so you might have to **market** it to them. Wow huh?

That's not the end. Let's add some more limitations to whatever we have so far. What about budget constraints? And resource constraints? No company in the world will allow you to operate on an unlimited budget or with infinite timelines. All the decisions that you have to make will have to be made **under constraints**, either cost-related, time-related, or resource-related ones.

And that should wrap everything that the perfect PM does in their lives. Of course, none of us are going to be perfect PMs anytime soon, and especially not during the internship. So let's stop thinking about the ideal scenario, and get back closer to reality, which leads me to the next section,

What companies are looking for

Now this is a tough question. What do companies want in their candidates? If that question was easy to answer, then everybody would be getting hired as a PM. But that doesn't happen and so what I can do here is just guess what companies are looking for, based on my and other people who got hired as PM interns' past experiences.

I think the one thing that every PM company is looking for in its candidates is **clear and highly-structured logical and decisive thinking while operating under several constraints.** I know that it is a lot of globe in one sentence, but it will have to do, because even if you ask companies what they are looking for, they will also give you globe answers.

Talking of companies, some of the PM companies that hired during the 2017 internship season were:

1) Microsoft
2) Amazon
3) Samsung
4) Visa
5) Sprinklr
6) Media.net

7) Infoedge
8) Zynga
9) Drivezy
10) Lenskart
11) HUL
12) P&G
13) People Interactive

During the 2018 season, the list changed quite a bit:

1) Microsoft
2) Amazon
3) Sprinklr
4) Media.net
5) Airtel
6) Vodafone
7) OLX
8) AppDynamics
9) Streamoid
10) MakeMyTrip
11) Dunzo
12) Grofers
13) Other companies like Zynga etc. that came but didn't hire anyone

The companies that come sell diverse products, ranging from Operating Systems (Microsoft) to mobile games (Zynga). Many companies also look for specific details in their candidates. For example, Amazon asks potential interns if they know about the 'Amazon Fly Wheel'. Microsoft asks potential candidates about some Microsoft product. Vodafone takes an online test before the interview.

But beneath these differences, they are looking for some base skills in their PM candidates. If we go back to our globe (and **bolded**) sentence, we can see that by focusing on the following areas:

1) Improving thinking method
2) Developing user focus
3) Improving communication skills
4) Improving understanding of technology

one should improve their chances of cracking the PM interview. Don't be overwhelmed now. The next few sections of the book should help you by demonstrating ways of thinking so that even you

can crack the globe.

A point that I want to make before closing this chapter is, almost all of the people who get a PM internship have some technical background. The background can either be a technical degree and/or a technical job. And in a way, taking candidates with a technical background makes sense for the companies because the products that these companies require you to manage will definitely involve technology of some sort, if not the latest then maybe some outdated one. And the assumption is that people with a technical background have handled technology in the past and thus can handle it again in the future.

What about work-experience requirements for PM roles? Tough to generalize here because companies also change their preferences. In 2016, Microsoft took IIMB interns without work-experience but in 2018 it didn't. But I guess having **relevant** work experience does help because there is a chance that in your previous work, you would have faced technical issues (that hopefully you didn't create), that you found solutions too. Again the assumption is that past experience will help in solving problems in the future.

But there are always exceptions. If you are a CA and want to try getting into a PM company, by all means go for it. The only caveat is that you will have to prepare harder than those non-CAs who made the mistake of studying for Engineering. Trust me, it can be done. Engineers are well known for making mistakes (see the crashing self-driving cars and failed satellite launches). Catch them when they do and beat them at their own game.

With the globe out of the way, let's start preparing for a role as a PM intern!

Chapter 3

How to begin Preparation

Some resume pointers

So you are ready to start preparing for the PM GDs and interviews? Hang on. Not so fast. There are lots of hygiene factors that you will have to take care off before beginning to prepare for the interviews. One of the most important hygiene factors is your 1-page resume.

"Oh. But the resume is only a 1-pager. Will it take long to make?"

Yes, the resume will take very long to prepare. That's why the exercise starts early in the first term and continues until the second term. Resume preparation takes time because you will have to draft points in the right manner and then get all of them approved by a third-party.

The resume-building process starts with you being assigned resume mentors. Depending on when you get hold of this book, you might or might not have received enough gyan about resume-building from your mentors. So let me also take this opportunity to pass on some resume preparation gyan. But before that, I want to add a disclaimer that the forthcoming tips are meant to be guides and not laws of nature. Adopt them if they make sense and fit the current context you are in.

Gyan 1: No one knows how to prepare the perfect resume. You might have already received a lot of tricks and tips to make the perfect resume. Some of them will be correct. Some of them might even be contradictory in nature:

Mentor A: Remove the buckets in your resume because they consume too much space. Add some more text

Mentor B: Put buckets in your resume because they bring out achievements

The truth is, no one really knows what companies look for in resumes, and so they guide you based on their personal experience, which might or might not be appropriate for you. Take all advice and follow the tips that suit your judgement

Gyan 2: Don't remove technical points from your resume. This is a common mistake that many people make, and in the case of a PM resume, this mistake shouldn't be made. If you have had a technical job in the past, bring it out in the resume. Don't remove the points thinking that you are in business school now and companies will only hire you if you have non-tech points in your resume. PM companies like to see tech on the resumes.

Gyan 3: Follow Gyan 2, but present the technical points in an appropriate manner. You have done

tech in the past. That's good. Put that in the resume, but only in a proper format. A point like this:

Designed satellite ground-station code using PIL, Numpy, Pandas, OpenCV Python3 libraries

Is good but not that helpful for a PM resume. Remember, the role you are applying for is NOT a coding role, and spelling out Python libraries or frameworks will only consume precious resume space and not convey much to the interviewer in terms of the impact of what you did. A better point might be:

Part of successful satellite launch, developer of spacecraft and ground station software

Why is it better? Because it removes the technicalities and makes the point more readable for a screener, who might not have a technical background. Also, it conveys some technical impact to a screener who is looking for someone to create a technical impact during the internship. The general advise here is that bring out the technical experience but present it in a way that shows the magnitude of the impact on the company you were working for (either on the company operations or on the financials). By writing your points in a way that show impact, it will convey to the screener that what you did in the past mattered to the organization.

Gyan 4: If you have dealt with any sort of new project/product in the past, highlight that as project/product management. Companies that come on campus will put you on a team that is managing a product. And management requires responsibility and decisive abilities. Thus, let that come out in your resume by showcasing appropriate past responsibilities.

In my work experience, I had managed a small product before, and I mentioned that in the resume. During the interview, they then asked me about that specific point, because it was highly appropriate for the role.

Gyan 5: Highly inflated numbers is not a must. As you go through the multiple iterations of making your resume, you might receive advice like ' without having inflated numbers the company will throw your resume out' etc. That's not true. Having inflated numbers in your resume might help, but not having them doesn't mean you are going to get rejected, because many companies know that the numbers on students' resumes are exaggerated. The biggest number I had in my resume was in terms of thousands of dollars. Not impressive by any scale if compared to the Crores that some people had mentioned.

In fact, inflated numbers might also turn out to be a curse if you are suddenly asked to explain during the interview exactly how you arrived at that number. For example, Amazon, during the interviews, did ask its potential hires how they arrived at some of the numbers mentioned in the resume.

Some general pointers

You are not ready to build the perfect resume. That's great. Let's move onto the other hygiene factors.

PPTs of PM Companies: If you don't know what a Pre-Placement Talk (PPT) is right now, you will find out soon enough. Several PM companies give PPTs before the placement week. How useful is it to attend these PPTs? In my view, moderately useful. Sometimes they talk about new products or technologies that the company is building. For those companies interviews, that might become an interview question. Some companies like Zynga even offer Pre-placement Interviews to candidates in their PPTs. For other companies the sessions are general company introduction and not overly helpful while preparing for a PM role. The best course of action for you can be to find out which company is giving the PPT and ask around whether the company does anything more than introductions in its PPT.

Case Studies/Design Competitions: Taking part in these competitions can be a good way for the company recruiters to get to know your ideas (in case they are involved in the judging process). And if you win, you might also get a PPI.

Completing PM tasks such as wireframe design development: Often during the course of online applications, many companies will open tasks that will require you to maybe make a mobile application using wireframes or other techniques. Some companies like Media.net require you to complete these tasks so that they can judge your thinking process before the interviews. Do look out for these activities to get a place in those companies interview panels.

Forming groups for interviews & GD prep: Many of us prepared for the interviews and GDs in groups, rather than individually. You can also try forming a group with similar people who are also looking to get into PM roles. Preparing in a group has several advantages:

1) Different group members bring with them diverse ways of solving a problem. Some of those solutions might be more efficient and optimized than your solution.

2) You get to hear their technical background and how they dealt with technical problems in the past. This knowledge can be useful in interviews if you are given a problem to solve that you have personally never encountered, but, by luck, you had talked to a person who had been in that similar situation before. You can mould your solution to the interviewer according to what you heard from someone else.

3) Tasks can be divided among each person in a group and conquered together. Preparing

for a PM role involves several tasks (a flavour of which you will get in the next chapters). Given the limited time and energy you have, you can divide the workload among the group. For example, one person can prepare daily technical news tidbits, while the other person can look through company websites and make a summary of all their products. With all bases covered, information can be shared amongst group members. The advantage here is that you will get all the necessary knowledge without having to spend an extra-ordinary amount of time in finding information.

4) You get motivation and encouragement. You might need external motivation, because after a few days of constant practise, those product design and case sessions can get boring and monotonous. The others in the group can help by keeping you on track and pushing you along as well.

Prep resources

With all the hygiene factors covered, let's start with going over a list of the topics you should prepare for:

1) Resume-based questions
2) HR questions
3) Guesstimates
4) Product Design/Modification questions
5) Tech questions including latest technology and news
6) Technical Cases
7) Coding/Algorithm questions
8) Company-specific products

And here is a list of useful resources to get started with the prep:

1) *'Cracking the PM interview' - Book by Gayle* => Reading this book will help you structure your preparation process. It covers the basics of most of the topics in the list above.

2) *'Case Interviews Cracked' - Book by Kelshikar & Garg* => This book is a case-prep book and contains a lot of example cases and their solutions. It provides a good brief about how to solve cases. And the knowledge and skills you learn from this book can be ported over to solving other PM type questions such as guesstimates, product design/modification questions, and technical cases. The goal of going through this entire book should be to learn how to structure your thinking in a way that it can be used to solve any given problem.

3) *TechCrunch and other technical news websites* => Useful for keeping abreast of the latest news from the tech-world

4) *Company websites* => Useful for finding out company-specific product information

Of course, this is not an exhaustive list of resources. There are so many more books and websites out there. Use them if you find them useful, but try not to get confused or overwhelmed from all the information present out there.

Does prep for other fields help in prep for a PM role?

This is a very frequent question that gets asked. To be honest, prep for Finance doesn't help in prepping for a PM role, because finance prep is knowledge-heavy and the knowledge that you will be learning for a finance role is mutually exclusive to a PM role. Prep for marketing helps a little because you might pickup some marketing frameworks while solving marketing cases. But again, marketing requires learning different skills, and those skills might not be directly applicable to a PM role.

On the other hand, prep for consulting does help. In fact, one of the resources mentioned above (Case Interviews Cracked) is a book that is often used while preparing for consulting cases. And many people used this book while preparing for PM roles as well. The reason why prep for consulting is useful while preparing for PM is that consulting prep teaches you how to structure your thoughts and break down problems to come up with the best solution despite a lack of information. And that is a very useful skill to have during a PM interview. Most of the time, the prime concern of the interviewer is to check how well you structure problems to come up with the best solution under given constraints.

Despite consulting prep being useful while preparing for a PM role, it is just a subset of the total prep required for a PM role, because a PM role also needs knowledge of technology and guesstimate-solving skills. So even if you do decide to prep for consulting, be prepared to put in extra effort to gain knowledge that is very specific to a PM role.

Chapter 4

Resume Based Questions

Finally, let's get to the serious stuff. The first type of question we are going to look at are resume-based questions.

What is your background? How much do you know about technology? Product Management is about solving customers' problems, and almost all of those problems are technical in nature. So a prerequisite to becoming a PM intern is to know about technology, and you will have to demonstrate that knowledge in some form during the interview or GD.

The first piece of information that a company gets about you is your resume. So it is necessary to work at preparing a good resume and once the resume is finalized, it is necessary to prepare answering any question that the interviewer raises on points in your resume. It is your resume, and you have spent the effort in writing it. So the interviewer will expect you to know each and every point in your resume. And to reinstate the point, if you have included numbers in your resume, be ready to explain to the interviewer how you arrived at the number. The larger the number, the more explanation from your side, it will need to convince the interviewer.

Some of the possible questions that can be asked on your resume are given below:

1) How did you arrive at this number?

2) So you have mentioned here that you have done project XX in the past. What was it about? What was the impact on the company's operations/revenue?

3) Walk me through your resume

I can't give you a right or perfect answer to the questions mentioned above. But I can say that in my case, a lot of my resume points are based on startups and international experiences, and so some of the questions I was asked were:

1) You have worked in small-companies your whole life. Was that a deliberate decision or something that you were forced to do?

2) Why choose a big company now?

3) What made you come to study in IIMB?

4) What were you doing in country XX?

I did try to link every answer to the questions above to the role I was interviewing for. For example, for the question, *Why choose a big company now?*, I remember mentioning something

along the following lines:

Me: Working in a small company is a very interesting experience but everything is on a small scale. A small market is targeted and customers are few. In a big company, I will be exposed to the same things that I did in a small company, but at a much larger scale…….

Yeah, I know, globe. As long as the globe manages to convince the interviewer, it's alright.

We talked about some tips to remember while making a PM resume. Here is another point that relates to the formatting of the points rather than the content. The suggested format for structuring points in your resume is as follows:

Power-Verb *(preferably bolded), followed by business impact and a* ***number*** *(preferably bolded again), followed by some technical detail*

Starting the point with a power verb helps to attract the reader to the point and shows action on your part. Stating a business impact shows how your work contributed to the business of the company, highlighting its importance in the larger scheme of things, and stating a number concretizes the business impact. Finally, the technical detail can highlight your technical expertise or also work as a filler to make the point fill the entire line.

Do remember to practise enough of these resume-based questions so that you can recite each project/responsibility/experience on your resume inside-out.

Chapter 5

HR Questions

HR questions, including behavioural questions, are those sets of general questions that might seem easy to answer at first glance, but can be deceptive because depending on how you answer, they can give away facets of your character and behaviour. HR questions are open-ended and can be answered in several ways. Thus they become good points to judge honesty, integrity, self-knowledge, existing biases, and other character-based traits you might engender. And with a lot of companies nowadays placing heavy premium on hiring candidates that 'fit the company culture', answering these questions well becomes as much of a necessity as of answering the other technical questions well.

Listed below are some often-heard HR questions:

Introduce yourself

Many companies that you interview with will ask this question right at the beginning. Even though they have your resume in front, they still want to hear about you from your own mouth.

Why?

I don't know. Maybe the way you answer the question serves as a truth filter because they can verify whether what you say matches what's written in your resume.

Does that mean I can answer the question in any way I like?

You can, but it's better to prepare a rough answer that makes a positive impact on the interviewer.
Whenever I was asked this question, I treated it as an opportunity to summarize my experience and signal that I was fit for the role being offered, and all within 2 minutes.

I kept the following things in mind:
- I have to talk about something unique I have done in the field the role was in
- I have to mention my experience/achievements in the field
- I have to give them the image that I am the right person for the job

When the interviewer at Microsoft, for the role of PM, asked me the question, I did a brief run-through of my job experiences, technical degree, and also talked about the time I had designed small-scale products for one of the companies I used to work for. I changed the reply when an Analytics company asked me the question. At that time, I talked about the large amounts of data I had handled in the past.

Now, specific for a PM role, covering these points should help create an impact on the

interviewer:

1) Technical projects or work done in the past => The idea behind mentioning your projects and experience is to highlight your familiarity with technology and assure the interviewer that you won't be overwhelmed when put in charge of a technical product.

2) A leadership role, or a product/project management role in the past => By your mentioning that you have already worked as a team leader or as a product manager highlights your fit for the role.

3) Some business impact => If you demonstrate that what you did in the past made a difference to the company, or some society in your college, it will show that you can do things that can create an impact, rather than blindly focusing on anything that comes along.

Why are you applying to this company

Companies ask this question to find out if you have done the pre-requisite research about the company and are able to articulate what makes it unique as compared to its competitors.

For this question, demonstrate that you did go the extra-mile to find out about the company, its products, and its culture. For example, if asked

Why do you want to join Microsoft?

You can reply something along the lines of:

I have been using Microsoft products since I was 4 years old, starting all the way from Windows 95 and Internet Explorer to Microsoft Office. Microsoft literally introduced me and my generation to the world of computing. For many of the people in my generation, the classic Windows symbol was one of the first company logo we remember seeing……

List three strengths

This is one of those questions on which textbooks can be written, because it is so commonplace. For this question, you can prepare a list of three strengths and then present those strengths in the context of a product management situation.

The book "Cracking the PM interview" has great examples of strengths that you can speak about during the interview. One of the examples presented in the book is:

"I have a ton of initiative. I'm not afraid to get my hands dirty or take a risk. At my last job, I felt a lot of non-engineers would benefit from understanding more about how the product is built. I started a weekly lunch program where engineers would give technical talks to non-technical people about the product, recent developments in technology, and other related topics. Almost everyone in the company came to at least one talk each month, and about 10 percent of the company attended every single talk."

See what the author did there? She put the strength in the context of a product management role, making the entire piece relevant to the interviewer.

List three weaknesses

This question is similar to the 'strengths' question above The interviewer wants to know whether you have insights into your own habits and character and whether you are openly able to vocalize them.

Here too, the book "Cracking the PM interview" has great examples of weaknesses that you can speak about during the interview. One of the examples presented in the book is:

"I'm often not very detail oriented. While this has some benefits, particularly in an entrepreneurial environment by helping me move faster, it also means I've made some embarrassing mistakes in the past. I've developed a good sense though of the types of situations where I'm likely to make mistakes, and I double or triple check my work in those cases."

Honesty about weaknesses should be key here.

If you didn't get selected, what will you do

The idea here is to check whether the internship is an end or a means to an end. Answers like

"I don't know what I'll do"
or
"I haven't decided"

clearly demonstrate that the internship is not part of a larger plan that you have with your career/life. On the other hand, answers like

"I'll continue learning about Product Management from other sources"
or

"I'll build my skills…."

demonstrate that you have a future plan in mind and that the internship is one of the tasks that you are doing to achieve that future plan. It shows clarity of thought and decisiveness.

Behavioural Questions

Similar to HR questions, behavioural questions can be very broad. Many-a-times, they are leading questions from your reply to some HR question.

Interviewer: What is one of your weaknesses?

You: Sometimes, I tend to skip over details. In my old job, there was an instance when I had to choose between implementing a project on time or implementing it accurately. Completing on-time would mean I would have to compromise on product-testing and release the software with potential bugs. Completing accurately would mean I would miss the deadline but the software would be bug-free. I chose the former option, foregoing some of the detailed product testing and code review.

Interviewer: There is an ethical dilemma here. Shipping a buggy product would affect customer experience. How did you resolve this dilemma?

You: I was aware of the dilemma, and in normal circumstances we might not have shipped a buggy product, but to the client, timing was more important than accuracy. So we were just following the client's immediate demand, thinking that future bug-fixes can address any customer complaint……..

As you can see, interviewers can use behavioural questions to **indirectly** judge your beliefs and values. They will be looking for the answers to some of these questions:

What would you choose between time and accuracy?
How inclusive and open-minded are you?
Would you favour your views over others' views?
How much do you value company culture?
Will you be a good fit in the company culture?

Using your answers they will determine whether you are the right candidate for the role. Answering these questions correctly is as important for you as answering the technical questions correctly, so do take time out and prepare for them.

Chapter 6

Guesstimates

Moving onto to technical questions, the first category of questions we will touch will be called guesstimates. Guesstimates are questions which require a number as an answer, but reaching that number from the question will require you to take knowledgeable guesses and use logical thinking. The question will often seem vague and unanswerable. Examples include "Calculate the weight of Mount Everest" or "How many police stations there are in India".

The idea behind these questions is not to come up with the correct number, but to break down the problem and solve it step-by-step so that you can come up to a reasonable guess of the actual answer. For example, if you are asked to calculate the weight of Mount Everest, you will have to break down the problem into several steps, and at each step along the way, make reasonable assumptions. By combining these reasonable assumptions, you can then come up to a final figure, which most likely won't be true, but will be an estimate of the actual value.

Why are these questions important? Because many times during your internship/jobs, you might be expected to take a decision, but full information is not available for you to make the correct decision.

Should I release this product now?
What questions should I include in customer surveys?
Should the marketing campaign touch upon these themes?

These are some of the questions you might need to answer but don't have proper information about. That's when having the ability to reasonably guess a correct answer becomes handy.

For you to get a better idea of what to do when a Guesstimate question does come up, let's solve an example question and come up to a **concrete answer**:

Question: What is the weight of Mount Everest?

Step 1: Breakdown the question and resolve any follow-up questions you encounter

Now you might be thinking, *What a stupid question? Who would want to know that? And more importantly, how would I go about solving such a question?*

The first step to solving this problem is to take the question, break it down word-by-word, and focus only on the words that are important. *What is the* ***weight*** *of* ***Mount Everest.***
The two bolded words are important. They mean that you need to come up with a kilogram estimate for the weight of the famous mountain.

Step 2: Define the scope and assumptions of your problem

Great, so you have to calculate the weight of the mountain. What next?

In my experience, there are only two methods of finding out the weight of something: Either you measure the weight using a weighing machine, or you use the ancient, yet trustworthy, Physics formula:

Density = Mass/Volume

You can't weight Mount Everest on a weighing machine, so that leaves us with the second option of finding the weight.

So you have determined that you need to find out the volume and density of the mountain. Since volume can only be found for regular shapes, you can assume that the mountain is a regular cone, and the volume of a cone is 1/3*Pi*Radius^2*Height.

So we have assumed the mountain to be a cone. What is the cone's angle? We can make the assumption here that the angle is 45°, based on the reasoning that the mountain has a very wide base (so that it can support such a large structure) with an angle of maybe 30°, and tourist photos show it has a steep top of maybe 60°. Averaging the two numbers out will give you a value of 45°.

What about the density? That will depend on the material that the mountain is made up off.
Here we will make an assumption that the mountain is made up of uniform rock with a single density. We know that there are other things such as trees and snow on the surface, but we are going to ignore those items, and calculate the weight of only the bare mountain.

Step 3: Form the equation

We have made the following assumptions in our quest to solve the problem:

1) The mountain is a regular cone with a volume calculated using the equation
   1/3*Pi*Radius^2*Height
   The cone's angle is 45°

2) The mountain is made of uniform rock throughout

Remember to openly state these assumptions and inform the interviewer that you are making these reasonable guesses based on the above-mentioned criteria

We are going to use the equation Mass = Density x Volume
Mass = Density x 1/3*Pi*Radius^2*Height

Pi * 1/3 is almost equal to 1
Height is almost 9000m
Radius is also almost 9000m (Using the 45° value for angle means the base of the triangle is going to be equal to the height.

Thus, $R^2H = 9000*9000*9000 = 7.3 \times 10^{11}$

Volume of the mountain: $7.3 \times 10^{11} m^3$

Density of normal rock = 3000 kg/m^3

Thus, the mass = $7.3 \times 10^{11} \times 3000 = 2.2 \times 10^{15}$ KG

Mount Everest weighs 2.2×10^{15} KG

Great. You have successfully calculated the weight of Mount Everest. A few points to note here:

1) The numbers taken might seem very crude, but remember that at that moment, you won't have access to the internet, and so all the numbers and equations you use have to be ones that you remember of the top of your mind

2) Explicitly state to the interviewer all the assumptions you are making throughout the calculation process. Solving guesstimates is not a test of accuracy.

3) What matters here is your process of solving the question, more than the actual answer itself. So even if the answer isn't completely correct, explicitly demonstrate your thinking process to the interviewer.

With these tips and tricks, practise solving as many guesstimates as you can so that the process of solving these vague questions becomes natural. To help you get started on your journey, here are some more practise questions:

1) Estimate the market size for Zomato in Bangalore
2) Estimate Blockchain market size in India
3) Number of Uber rides taken in Bangalore in a month

Many guesstimate questions will require you to make assumptions about certain demographic factors in India. Below is a list of some very common ones prepared by Sai Sandeep Yelchuri while we were practicing for our internships:

India Demographic Statistics

Population

Total Population= 125 Crores
Bangalore Population= 1.25 crores (1% of India)
Rural Population= 70%
Urban Population= 30%

Age Wise Divide

Average age= 28 years
<25= 50%
<35= 65%

Education Level

Literacy in India= 74%
Urban= 81%
Rural= 71%

Gender

Male : Female= 1:1

Average Life Expectancy

68 years

Income

Average Income per individual= 10000 per month.
This is largely pulled down by the very less rural income

Population Density

380/Sq. Km

Mobile Penetration

Number of Mobile Users = 60Crores
Number of smart phone users = 30Crores

Internet penetration

Total Penetration = 48 Crores
Urban = 65%
Rural = 20%

Automobiles

2 Wheelers = 12 Crores
4 Wheelers = 17 Crores

Credits: Sai Sandeep Yelchuri

Chapter 7

Product Design/Modification Questions

Probably the most important set of questions to prepare for your GDs and interviews, product design and modification questions are guaranteed to be asked during either GDs or Interviews or during both. That's because while interning as a PM, you will be given the task of designing a new feature or product or maybe improve an existing product. And to gauge whether you have that ability in you, the interviewer asks you to design or improve a product and judge your thinking process.

Similar to the case with guesstimates, the objective of the interviewer for asking these questions is to see whether you can think in a structured manner and with decisive clarity. With limited information available during the decision making process, you will have to make assumptions about several things. Do remember to state your assumptions to the interviewer clearly. Let them know how you are thinking.

For this chapter, I am going to differentiate between new product design and product improvement because they require slightly different frameworks.

New Product Design

So you are tasked with designing a new product. How do you go about it? What items should you consider while building the product? How will you measure success? The framework below should help you to understand what is required and come up with a product.

Thoroughly understand the problem: The problem provided to you will generally be in the form of a question. Some questions can be very vague, such as 'Design a clock', while others can be very specific, such as 'Design a clock for blind people'. You might think that the more vague the question, the easier it is to solve. In reality, it is the opposite. The less vague the question, the easier it is to solve because understanding vague questions takes more effort from your side.

If you vaguely are asked to 'Design a clock', you will have to ask the following questions just to understand what the problem means:

There are many types of clocks. Which clock is the interviewer referring to? Alarm clock? Grandfather clock? Souvenir clock?
Who is going to use the clock?

You will have to explicitly ask the interviewer questions to get the same amount of information that is provided in a specific problem statement like 'Design an alarm clock for blind people'

Do get as much information here as you can because only after thoroughly understanding what needs to be solved will you be able to design a correct solution.

Identify Users & Customers: Every product is designed to be sold. When the problem statement given to you specifies a product to be designed, there is already a target user for the product in the mind of the interviewer. Two scenarios can play out here. Either the interviewer provides the user segment to you, or the interviewer asks you to determine the right user segment.

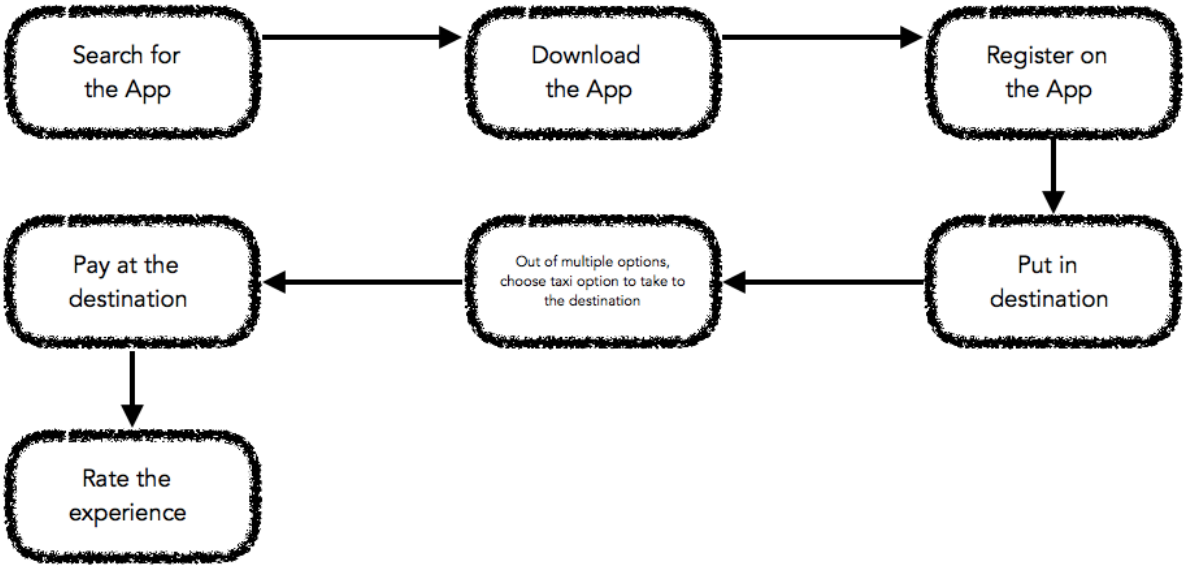
For very specific questions, such as 'Design an alarm clock for blind people', the user is already defined as blind people. For vague questions, ask the interviewer about this information, and if the interviewer declines to give it to you, you make an assumption, state it out loud, and stick to that assumption throughout the product designing process. You can say something along the lines of:

I'm going to design a souvenir clock for tourists to take back to their home countries. Incoming tourists are my target market.

Sometimes, the users of a product and the customers can be different people. The difference between users and customers is that users use the product, while customers buy the product. And this difference is usually present in products that are being designed for kids to use but for parents/teachers to buy. Do find out from the interviewer if that is the case for the product you are designing.

Understand the user journey: By now you thoroughly understand the problem and know who the target users of the product is. You are ready to study existing products in the market and how the user interacts with the existing products. Identifying the user journey will help you to understand the usage of the product.

For example, if the question asks you to 'Design Uber for kids', you will have to study the current Uber application and how users are using it right now to solve their needs. The chart below highlights the user journey of using Uber:



Identify pain points/gaps in the existing product: You now know how the user currently uses the product. There are obviously some gaps in the existing solution, and that is why you are being asked to design a new product. If you are being asked to design an alarm clock for blind people, you know that some of the pain points for blind people currently using the product are:

1) Brail-less buttons on the clock
2) No talking feature
3) Complicated operating interface (multiple buttons etc.)

In the product you are asked to design, identify as many pain points as possible so that you can show the interviewer that you are thinking through the entire user experience. Also, the more pain-points you identify, the more solutions you can generate.

Prioritize pain-points: You now have a list of pain-points in the existing solutions in the market. Prioritize the list according to what you think is the most important problem to solve. There are some pain-points which are a must solve, while some are just good-to-solve. Identify which pain-point is which. Use this list to define your product goals.

What features should my solution?
How do I generate solutions to match my product goals?

Generate solutions: You now have the list of product goals sorted from high priority to low priority. Start generating solutions for the problems and while you are generating the solutions,

always keep in mind that all of them are supposed to focus on the customer's needs rather than yours. Remember, it is always CUSTOMER FIRST.

Continuing the example of the alarm clock for the blind, some of the solutions that will solve the problem of the target user are:

1) The clock can be controlled by voice commands. When the clock is being configured for the first time, it can be given a name so that the blind person can address it without having to come near the clock at all times. The clock can also be in an eternal listening mode, and always listening for its name. As soon as the clock hears its name, it will get activated

2) Since voice operations consume more power than non-voice operations, the average power consumed by the clock will rise, making it necessary to have rechargeable lithium-ion batteries powering the clock. There should also be an option to power the clock from a wall-socket

These are just examples of solutions that can be generated. You can think of even more than these. A tactic here is to generate as many solutions as possible, and showcase them all to the interviewer. Ask him/her which one they prefer, so that you get a sense of direction to take for the rest of the problem-solving process.

Evaluate solutions based on relevant parameters: You have a list of solutions. You might have generated more solutions than needed for the problem. That means you now have to evaluate all the solutions based on some success parameter you define so that you can filter the solutions and choose the best among them. State to your interviewer what your assumed success parameter is. Finalize on one overall solution.

Pricing Model: You have devised the final solution and have explained it to the interviewer. Depending on the company and the question, the interviewer might ask you to come up with a pricing model for your solution.

Pricing model varies depending on the solution. For hardware products (like alarm clocks, pens, etc.), the pricing model can be a markup or margin based model on the cost of the product. Or it can be competitor-based pricing, in which you price the product similar to how competitors are pricing the product.

For non-hardware based products, the pricing model can be ad-based, periodic subscription based, commission per transaction based, or a one-time purchase.

Go-to-market: In case the interview reaches until here, you will have to evaluate the best way of

making the product reach the market. Will it be through advertising? Where and how often will you advertise? What about word-of-mouth advertising? Maybe you can give out free samples in the start to enthusiastic users and rely on them to use the product and spread a positive word about it to other potential users.

Success Metrics: You have made the product and it is now out in the market. How do you know you did a good job? How do you measure success? You will have to define some success metrics in the start that you can use to measure the success of your product.

Success metrics will be specific to products. For hardware products, success is typically measured by revenue earned in time period and whether that revenue is increasing from period to period or decreasing. For non-hardware products, success metrics can include downloads per period, ads revenue per period, user registrations per period etc.

I want to mention a closing pointer here about defining absolute versus relative metrics. Absolute metrics measure the metric in a certain time period, for example downloads per week. Relative metrics measure the same metric over time. For example whether downloads per week have been increasing or decreasing over the past year. Absolute metrics are useful comparison points across competing products. Which application has more downloads per week? Ola or Uber? And relative metrics are useful comparison points for the same product over time. Is Uber seeing increasing downloads per week over the past year or is it seeing a decreasing trend? Those data points can point to positive or negative trends about product adoption.

An example

With the framework in place, let's follow it with an example. Do remember to **constantly state your thoughts and assumptions aloud for your interviewer to hear your thought process. After all, that is what the interviewer is judging.**

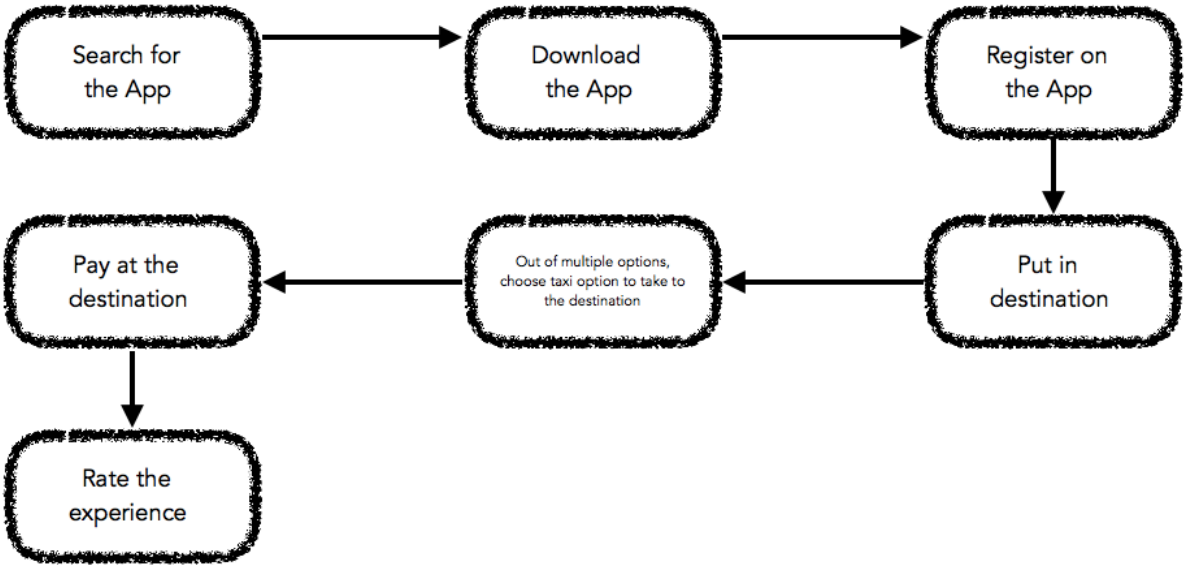
Problem Statement: Design Uber for Kids

Thoroughly understand the problem: What does the phrase 'for kids' mean? Does it mean that Uber should be usable by kids or Uber should have special provisions when the rider is a kid? This can be a real problem because nowadays, due to the convenience that Uber provides, many parents are opting to send their kids to school using Uber taxis. If the rider is a kid, does the question ask you to design the app for sending the kid to school or anywhere?

Ask these questions to the interviewer and get their perspective. For the sake of this problem, let's assume that the question asks you to design the app for the case that the rider is a kid and the user is the parent. The purpose of the app is to send the kid only to school and not any other location.

Identify Users & Customers: The question specifies that the users are kids. The customers will be the kid's parents.

Understand the user journey: The current user journey of the Uber application is as shown below. Do remember to physically draw out the journey and explicitly mention each step to the interviewer. Let them know that you are thinking through the entire product cycle from start to finish.



Identify pain points/gaps in the existing product: Several gaps exist in the existing user journey:

1) When registering your number on the application, there is no provision for registering a dependent's phone number

2) When sending kids to school, the security of the kid is of prime importance. There is no special provision for that

3) Since the user and the customer are different people, each ride will involve two parties, and both the parties will need to be connected to each other at all times while the ride is going on

4) The payment and rating has to be done by someone who is not the user i.e. the parent and not the kid

Prioritize pain-points: The pain-points above are in order of priority. They can be converted into

the following product goals:

1) Register a dependent's phone number

2) Have special security provisions for dependents such as kids

3) Ensure that the kid and the taxi are constantly in touch with the parent

Generate solutions:

1) While registering on the app, a question should be asked if the registering user is a parent. If the user selects 'Yes', then the child's phone number (if he has a phone) can also be asked.

2) While using the app to send the kid to school, when the parents puts the school as the destination, a pop-up should come that asks the question if the rider is going to be a kid.

3) Live-tracking of car's location constantly being shared with the parent

4) While the ride is going on, any abnormal stop the taxi makes is flagged in the application and the parent gets a notification

5) The time to school from home is pre-calculated before. As soon as the ETD is exceeded for the journey, if the child has a phone, the child will get a call every 2 minutes. Otherwise the driver will get a call from the Uber call centers every 2 minutes asking what the delay is and the current status of the ride.

6) While searching for drivers, Uber will prioritize finding women drivers

Pricing Model: Uber for kids will follow the same business model as the regular Uber does, but the prices will be hiked, maybe by 15%, due to the extra security services that Uber has to offer.

Go-to-market: Uber grew in size primarily due to word-of-mouth. Since Uber is already an established company, the added feature will reach Uber's current audience through the application. Maybe Uber can show banners at the start of the app that highlights the new features offered.

Success Metrics: How to measure the success of the solution? Relative metrics such as the following can be used:

1) Number of registered kids per week

2) Trips to schools made per week
3) Security services offered by Uber used each week

These weekly numbers can be compared over multiple weeks to determine whether the trend is rising, constant, or falling. The result of that observation will determine whether the service is a success or not.

After you do the exercise, and in case the interviewer asks you to give a one minute summary of the entire solution, you can say something along these lines:

Uber for kids will be similar to the regular Uber service with an increased focus on security of the kids. From the current user journey, we can see that pain-points exist during finding the driver and completing the ride. While finding the ride, Uber will prioritize finding women drivers. If it can't find women drivers, it will prioritize finding experienced and highly-rated drivers (because the assumption is that experienced and highly-rated drivers care more about completing rides perfectly and so will be trustworthy for kids. Other safety precautions will also be taken by Uber such as showing the current GPS reading to the parent, allowing the parent to call the driver to talk to the kid at any time, warn the parent in case the car stops somewhere for an inordinate amount of time, and also contact the driver, asking for status, in case the actual time of the ride starts exceeding ETD. To cover for the extra costs that Uber is going to incur, a 15% margin markup is going to be charged to the end-customer. In terms of success metrics, the following three relative metrics will be used to measure success of the service: 1) Number of registered kids per week 2) Trips to schools made per week 3) Security services offered by Uber used each week. These metrics will be measured over multiple weeks to determine whether the trend is rising, constant, or falling. Using that information we can declare whether the service is a success or not.

Product Modification

What if the question posed requires you to modify an existing product? How would you go about that? The structure to be followed is similar to the structure that you followed while developing a new product, but with a variation in the way you start solving the problem:

Understand what the goal of the modification is: Why is the interviewer asking you to modify the product? The interviewer should specify their goal in the question itself. For example, if the interviewer asks you *"How can Internet Explorer be made more appealing to users"*, the goal is directly stated: Make Internet Explorer more appealing.

Upon hearing the question, you can ask the interviewer a further question: *"What does appealing mean? Does it mean to increase revenue for Microsoft or to get more users on board or to retain the existing users?"*

If the interviewer replies that the goal is to *"improve user experience so that new customers use the browser and old customers don't leave it"*, then that concretizes the problem being asked. You now have a concrete aim while trying to improve the product:

Improve the user experience of Internet Explorer to attract new customers and retain old customers

Identify the product and its existing problems: You now have a goal in mind. How are you going to begin improving the product? Do you know the problems that the current product faces?

Taking forward the example from the previous section, the product we have to improve is Internet Explorer. What are the existing problems in the product? Some of them that I can personally identify from the time I used to use the product are:

1) Slow browser operating speed
2) Poor UI

What if I have never used the product in question? Inform the interviewer that you have never used the product and can't identify its problems. Either they will change the product in question or they might even tell you what customers sight as being the biggest issues with the product.

Generate solutions: The idea here again is to not come up with the perfect solution, but just a good enough solution to demonstrate structured thinking. In our case, how do you solve browser operating speed issues?

A possible solution is that you first identify the bottlenecks in the data flow process from back-end to front-end, and try to optimize the bottlenecks:

"Look at the browser code and open the part which displays the front-end UI by right-clicking on the UI and choosing the option of seeing page source code. Note down all the procedural calls that are being made and time their operations, whether the operations deal with the backend database or third-party locations or any other locations. From all the measurements, identify the queries that are taking the longest to execute. Optimize that query and do it for other queries that are taking too long to execute."

Success Metrics: How will you measure whether what you have proposed will result in success? Some of the success metrics you can use are:

1) Number of users using the product per week
2) Time spent per user per week

3) Feedback from users using a feedback system (if there is one)
4) Specifically for measuring the performance of browser speed, you can use metrics like page-load times and content load-times. These metrics can be compared to metrics measured from other browsers.

Again relative metrics should be used and the metrics should be compared over several weeks to get an idea of the trends and looking at the trends determine whether the suggestion is a success.

Chapter 8

Tech Questions

The tech questions asked by the interviewer can vary widely in scope from awareness about the latest happenings in the tech world to knowledge about technology that is in the process of being adopted. The idea behind these questions is to test your general knowledge and awareness of events in the technical world.

These questions get asked both in GDs and in interviews. In GDs, the questions can be framed like this:

Talk about the new GDPR law enacted in the EU. What are its implications?

or

Augmented Reality is changing the way business is done. Evaluate the sentence

To answer generic questions like these, you will have to have some knowledge about the following technologies:

1) IoT
2) Cloud Computing
3) AR/VR/MR
4) Machine Learning
5) Smart-cities applications
6) Blockchain

And about the following policies:

1) Green technologies
2) Privacy Laws
3) Technologies being adopted by government for administration purposes

Note that the list above is not comprehensive, and it will change as technologies gain favour or fall out of favour.

Another useful division that technology is divided into nowadays is in the form of which field it will be used in:

1) Financial Technology (FinTech)
2) Educational Technology (EduTech)
3) Medical Technology (MedTech)
4) Others

A GD topic such as *'Blockchain is being heavily adopted across the financial sector'* will require you to to be aware of happenings in the Fintech world.

How much do I have to prepare?

That depends on your background. If you are a technical person and have been dealing with technology for a few years now, preparing for the above-mentioned topics will be relatively easier as compared to a person who is not from a technical background. You don't need to read what 'Machine Learning' is or what 'Cloud Computing' means. You just need to brush up on the latest happenings in the tech-world, and for that I used websites like:

1) www.techcrunch.com
2) www.engadget.com
3) Your favourite Tech-News website

In case you have no background in technology, then you will have to read up on the basics of the several types of trending technologies and get an idea of how they operate. One way to quickly learn the basics of trending technologies is to read the book *'Management Information Systems (MIS) in Business, Government, and Society'* by Rahul De (IIMB Professor). Depending on the edition you get a hold of, you can read the chapters that relate to the technologies listed above, and it will provide a decent grounding in technology. Supplement that reading with the latest technical news from some of the websites mentioned above, and you are ready to rock any tech questions that come your way in the GD or interview.

Chapter 9

Technical Cases

In this chapter, we deal with solving technical cases. Though not strictly needed for PM roles, and more widely tested for roles in tech-consulting companies like PWC Tech or Deloitte, technical cases are good to solve because they help in building the required structured thinking that is necessary for solving other PM questions.

The aim of a technical case is similar to that of a new product development question and even a guesstimate. You are given a problem and are required to come up with a solution in the most structured way possible. While you are finding the solution, the interviewer is looking to judge your thinking process. Thus, even in this case, do loudly state your thoughts and everything that goes through your mind on your way to find the right solution.

Let's try solving a technical case in a structured way:

Interviewer: Over the past few years, Company X, a product firm, has been facing a loss in business. Investigate why and come up with some recommendations based on the investigations

You: So the exact problem statement is '*Over the past few years, Company X, a product firm, has been facing a loss in business. Investigate why and come up with some recommendations based on the investigations*'

I'd like to ask you some clarifying questions here so that I can understand the problem better:

1) *How many years is 'past few years'? Is it just 2/3 or more?*
2) *Does company X have several products? What are the businesses like for individual products?*
3) *What does loss in business mean? Does it mean a loss in revenue/profit/number of customers?*
4) *Is the loss in business consistent over the past few years? Or does business sometimes come up while sometimes goes down?*

Interviewer: You can assume that the business has been consistently making a loss for the past three years. By a loss in business, I mean a decrease in the profit. X does have several products but only product Y has been registering a fall in profit.

You: So to summarize the problem statement, over the past three years, Product Y at Company X has been registering a fall in profit consistently. However, before that it was profitable. Is that correct?

Interviewer: Yes

You: So the problem is that recently profits have started falling. Now profits fall for two reasons,

either revenue is going down while costs stay the same, or costs are coming up while revenue stays the same. Do we have data to show which case it is in the case of Y?

Interviewer: Let's assume that its revenues are going down while costs have stayed the same

You: Ok. So the problem is with revenues coming down. Now revenue is calculated by multiplying the price of the product with the quantity of product sold. That implies that with revenues coming down, either the price of the product is coming down or the overall quantity being sold is coming down. Which case can I assume in this situation?

Interviewer: Let's go with overall quantity coming down. The price of the product has been the same relative to the past years.

You: So the overall quantity being sold is coming down. Overall quantity being sold is measured by multiplying the number of customers with the average quantity bought by each customer. Can I assume that in this case, the number of customers is coming down while the average quantity being bought by each customer is the same?

Interviewer: Sure

You: So the number of customers buying Y has fallen over the last three years. We have to determine why the number of customers has fallen. I'd like to ask, what is the specific customer segment that Y serves to?

Interviewer: So Y is a common product used inside IIMB. There are two segments it serves: 1) Students 2) Teachers

You: And any data to show which segment has been falling?

Interviewer: Data shows that the number of student customers has been the same over the past years but the number of teacher customers has been falling

You: So the number of teacher customers has been falling. It might be a possibility that the number of teachers on the campus has been falling, but I'll assume that the reason for falling number of teacher customers is something more significant. Now customers would use Y if the customers have a pressing need to solve and Y solves the need better than its competitors do. Thus, for product Y to sell, the following conditions have to exist:

1) *Customers have pressing needs*
2) *Y solves needs better than competitors do*

If any one of these factors is affected beyond tolerable limits, Y's sales will be affected.

Interviewer: Nice points. There have been reports recently of a new company that was incubated in the IIMB campus that was able to solve the customer's needs at a cheaper price.

You: So can we zoom the problem down to the fact that due to increasing price competition from a small company incubated in NSRCEL, Y has seen a decreasing number of sales?

Interviewer: Sure. Let's assume that's the reason. Now the recommendations?

You: I can think of a few, but they are vague since I don't know what the product actually is or which industry the company operates in. Some of the tactics that Company X can employ to sell Y better are the following:

1) *Increase barriers-to-entry for the new company (from Porter's 5 forces) by maybe certifying the product or emphasizing the importance of costly privacy/security which Y offers but the new competitor does not*
2) *Decrease costs by utilizing either economies-of-scale by doing more research to find out how production can be done cheaper. Maybe they can just outsource it to a cheaper location*
3) *Change the company strategy by focusing more on quality rather than price and emphasizing the factor to your customers*
4) *More marketing campaigns*

Interviewer: Good. Thanks for the solution. Let's wrap it up here.

As you can see, solving cases is an exercise in logical and step-wise thinking through each scenario. As mentioned previously, it is a good habit to keep talking about what's going through your mind while you are solving the problem so that the interviewer gets a clear idea of how you think, which is the item that is being evaluated. In the end, the approach to solving the problem matters more than the solution that you arrive.

Chapter 10

Coding & Algorithm Questions

A product manager role should ideally not involve coding, nevertheless, it is good to know some basic algorithms and practise these questions because solving algorithmic questions is similar to solving cases: You are given a problem and you have to find a solution to the problem, and most of the time you have to find the most optimal solution to the problem.

To practise algorithmic questions, you can focus on a few specific areas:

1) Algorithm complexity (Big-O Notation)
2) Data Structures
3) Searching algorithms
4) Sorting algorithms

Big-O Notation

The Big-O notation is used to express algorithm complexity in a simplified notation. Using an algorithm's Big-O value, you can determine how the algorithm will scale as the number of inputs to the algorithm increases. The lower the Big-O value of an algorithm, the more scalable it is and lesser resources it will need.

The most common Big-O notations are:

1) O(1) => The time taken to run through the algorithm is independent of the number of items in the array. Also called constant-time algorithm.
2) O(log n) => Time taken is proportional to the log of the number of items in the array.
3) O(n) => Time taken is proportional to number of items
4) O(n log n) => Time taken is proportional to number of items x log of number of items
5) O(n^2) => Time taken is proportional to number of items squared.

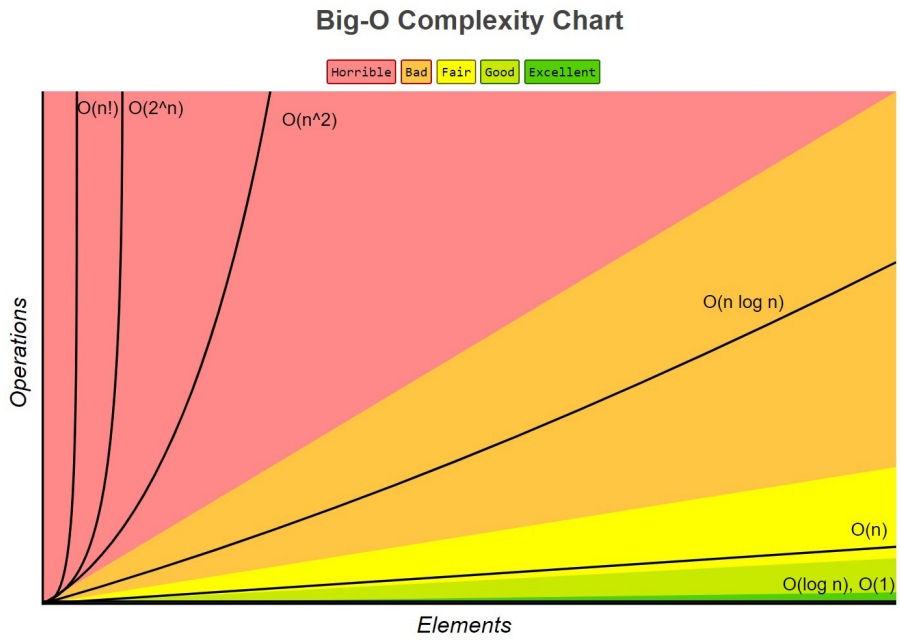


Image Source: https://cdn-images-1.medium.com/max/1600/1*KfZYFUT2OKfjekJlCeYvuQ.jpeg

Data Structures

As the name implies, data structures are algorithmic structures for storing and handling data. The most common ones are:

1) Array

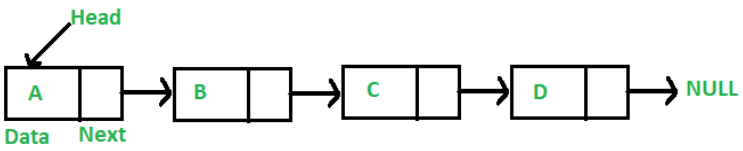
The most popular and common data structure, arrays are single-dimensional lists of elements. Elements (alphanumeric/memory addresses etc.) can be addressed by calling the 'index' of the array. Array indices start from 0.

2) Hashtables

A structure which stores data in the form of key-value pairs. A 'key' is often a unique value representing something about an object, while a value represents characteristics of the object. For example, in a library, books are given unique numbers. These numbers can become 'keys' in the hashtable. Furthermore, each book has several characteristics such as Name, Author, Publishing House etc. which can become 'values' associated with that 'key'. To look up information about the book, search for the serial number of the book in the hashtable and read the associated values with the key.

3) Linked-List

A structure with data-nodes linked to each other using pointers. Each node has a data field and a pointer field that points to the address of the next node. Linked-lists seem to be more complicated than arrays even though both of them serve the same purpose, but their advantage is that inserting/deleting nodes in a linked-list is easier as compared to doing that in other data structures. Also, linked-lists are dynamic data-structures, meaning that nodes can be added/removed even after the structure is declared, and unlike static structures like arrays that have to be pre-defined with a fixed size.



4) Stacks

A stack is like an array, but with limited operations that can be performed on it. Stacks are LIFO (Last In First Out) structures, meaning that the last element added to the stack will be the first element popped out. Only two operations can be carried out on a stack: push, pop. Pushing involves pushing elements onto the top of the stack, and popping involves taking out elements from the top of the stack.

5) Queues

Queues are similar to stacks, but they operate on a FIFO (First In First Out) basis. Similar to stacks, only two operations can be performed on queues: enqueueing, dequeueing. Enqueueing means inserting elements on the back of the queue. Dequeueing means removing elements from the front of the queue.

6) Trees

A tree is a structure that is a collection of nodes, but arranged in the form of a tree, with each node spawning a set of children nodes, and those children nodes spawning their own set of children nodes. The diagram below shows an example of a tree with multiple hierarchies of nodes.

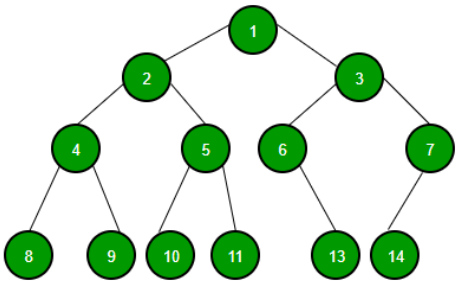


Image Source: https://www.cdn.geeksforgeeks.org/wp-content/uploads/binary-tree-to-DLL.png

Searching Algorithms

Search algorithms are used to search for data in any data structure. Out of all the existent searching algorithms, the most prominent ones are:

1) Binary Search

This algorithm is used to search for a specific term in a sorted data structure, usually an array. The algorithm starts by taking the query to be searched, and comparing it to the middle value of the array. If the query is larger than the middle value, it means that the value has to be in the right half of the array. If the query is smaller than the middle value, it means that the value has to be in the left half of the array. Once the comparison between the desired value and the middle value is made, the other half of the array is discarded. These steps continue until the desired value is found. If it is not found, a null can be returned.

Thus in each run, half of the array is discarded, bringing the complexity of the array to a worst-case scenario of O(log n). The photo below represents the simplified operation of the algorithm.

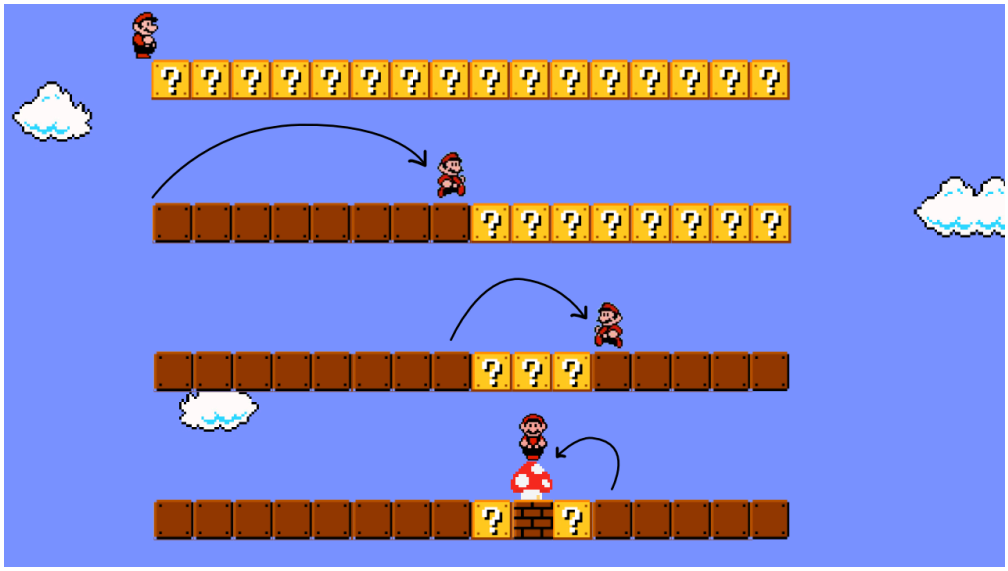
Source: https://www.topcoder.com/wp-content/uploads/2017/07/binary-search.png

2) Linear Search

Another search algorithm that simply involves searching for the desired element through an array by going through each element of the array one-by-one until the element is found. Thus, the worst-case time complexity of a linear search algorithm is O(n).

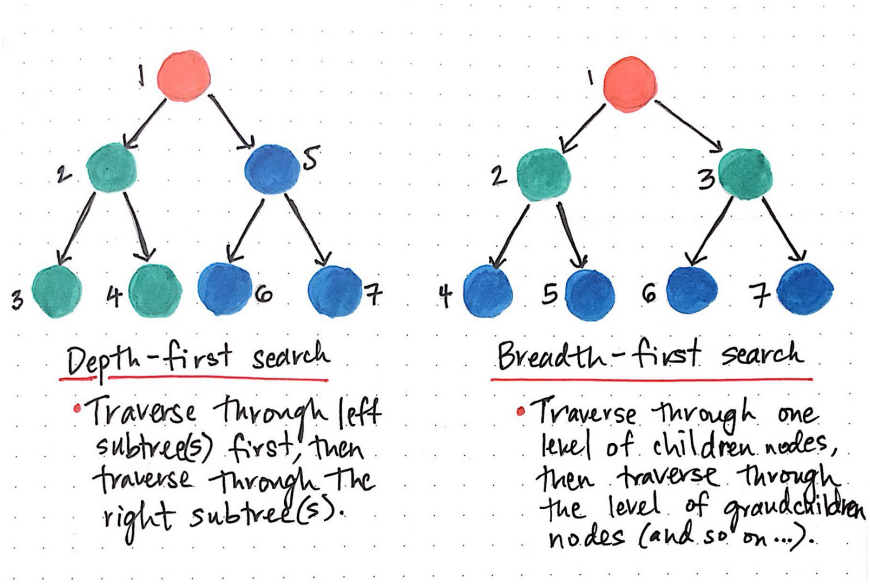
3) Basics of Breadth-First Search and Depth-First Search

These algorithms might not be necessary to understand, but I guess it's still good to know what these terms mean. Breadth-First Search (BFS) and Depth-First Search (DFS) are more complicated search algorithms done on graph data-structures. In a BFS implementation, the graph is searched



horizontally, covering the entire breadth of the graph. If the desired term is not found, the algorithm begins searching the tree one level below what it was searching in previously.

In a DFS implementation, the graph is searched vertically, by searching a node, and then its child node, and then that child nodes child node, until the search algorithm reaches the bottom of the graph. If the desired term is not found, the search goes back to the top and starts again adjacent to the node on which it began search in the previous iteration. The diagrams below hopefully give you a better idea of how BFS and DFS work.



Source: https://cdn-images-1.medium.com/max/1600/1*VM84VPcCQe0gSy44l9S5yA.jpeg

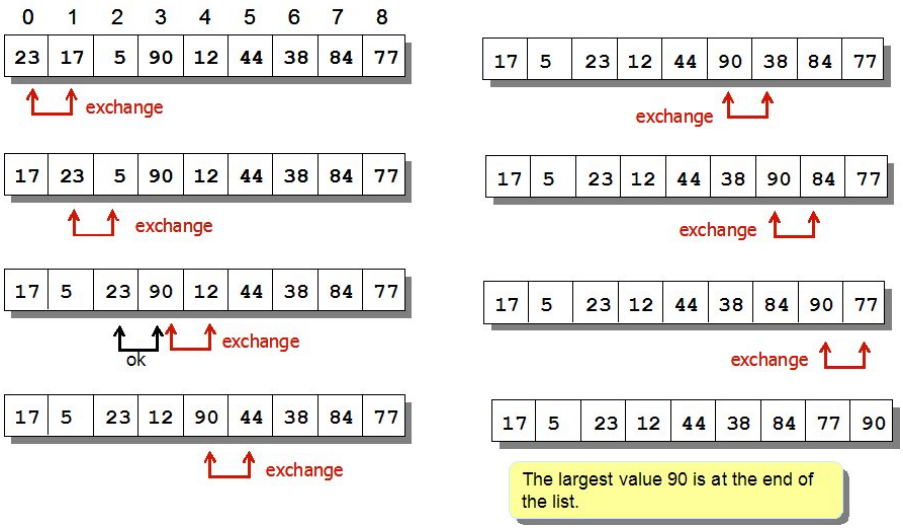
For these two algorithms, the Big-O notation can't be used because the notation doesn't work on graph data-structures.

Sorting Algorithms

Out of the several sorting algorithms, the most well-known ones are:

1) Bubble Sort

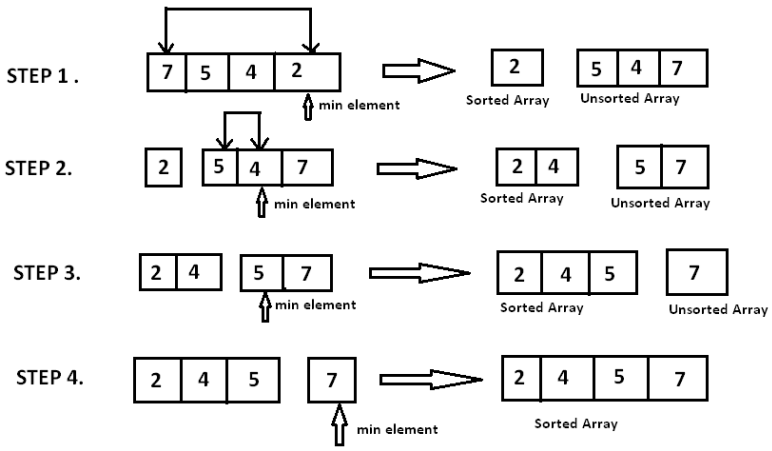
The bubble sort algorithm begins at the start of an array, and compares the value in that cell to the cell in front. If the value is smaller, it will swap the two cells. After each swap, the algorithm will move forward by one cell. One iteration of the algorithm will be over when the last element in the array is reached. If the array is not sorted then, another iteration will begin from the first value again. This process goes on until the entire array is sorted either in an ascending or descending order. Due to the algorithm having to move through the array multiple times, the worst-case time complexity of the swap is $O(n^2)$. The image below will help explain the algorithm more clearly:



Source: https://codemanic.files.wordpress.com/2014/03/bubblesortinteration.jpg

2) Selection Sort

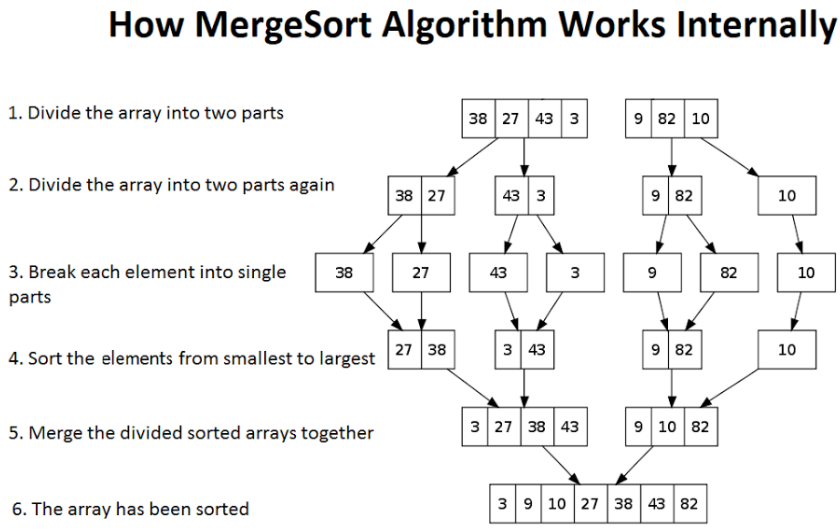
In a selection sort algorithm, the array is traversed to find the smallest element. Once that element is located, it is switched with the first element of the array. The algorithm then goes through the array again to find the second smallest element in the array and switches that value with the second element in the array. This process repeats until the entire array is sorted. Because the algorithm has to go through an array multiple times, the worst-case time complexity of this algorithm is also $O(n^2)$.



Source: https://he-s3.s3.amazonaws.com/media/uploads/2888f5b.png

3) Merge Sort

A powerful sorting algorithm, Merge Sort uses the 'divide and conquer' mechanism to recursively break down the array into smaller sub-arrays until only two element are left, and then sort the arrays on the way to assembling them back into a whole array again. Due to this recursive division, the worst-case time complexity of the algorithm becomes O(n log n). The image that represents the working of the algorithm is shown below:

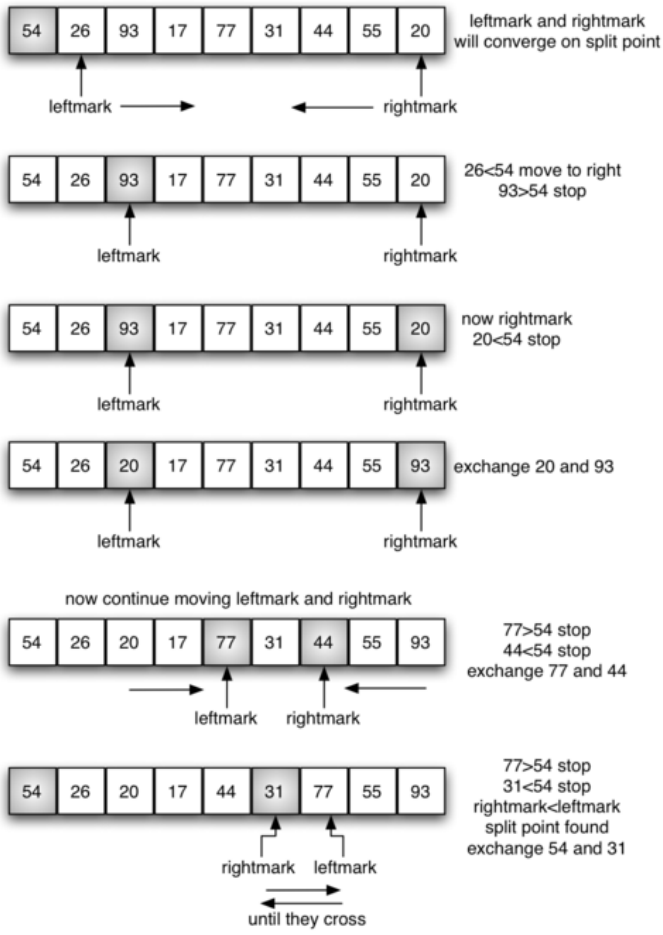


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4) Quick Sort

Quick Sort is similar to Merge Sort in a way that it also relies on recursion to sort the array. The difference is that quick sort relies on a pivot cell, and the sorting happens around that number.

Two pointers (one at the left and one at the right) continuously move, as shown in the diagram below.

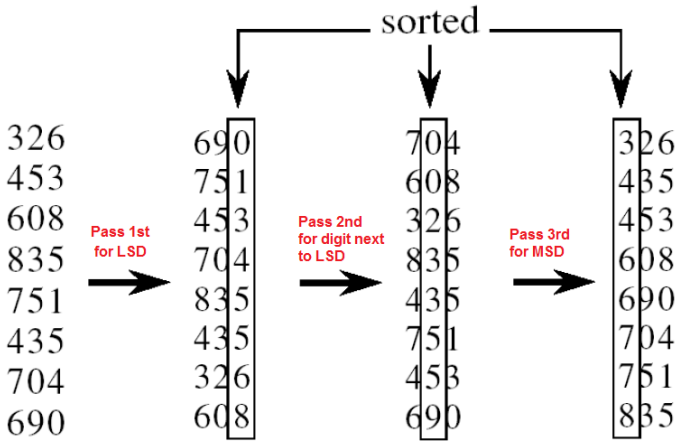


Source: http://interactivepython.org/courselib/static/pythonds/_images/partitionA.png

This step occurs recursively, until the entire array is sorted. Since several array traversals are required for the sort, the time-complexity of the algorithm is O(n log n), but in the worst-case, it can become $O(n^2)$.

5) Radix Sort

Radix sort operates differently from the rest of the sorting algorithms. It takes in a list of numbers, and sorts the numbers based on the Least Significant Digit (LSD), continuously moving from the 1's place to the 10's and then the 100's until how many ever digits the biggest number in the array has. This makes its worst-case time complexity O(nk), with k being the number of digits in the largest number of the array.



Source: https://www.codingeek.com/wp-content/uploads/2017/02/radix.png

Chapter 11

Company-Specific Products

Finally, we reach the last chapter of preparation of how to prep for a PM role. The goal of this chapter is to give insights on how to prepare for specific companies that come onto campus. Some companies, such as Microsoft expect you to know at least a bit about the different Microsoft products that are in existence. Other's like Sprinklr also expect you to know what they do and how they make money. Amazon expects you to know about its principles and the Amazon Wheel.

One way to go about preparing for specific companies is to get to know the companies that are coming for the PM role, if not from the Placecom team then from seniors or super-seniors, you can get to know at least most of them that are coming. For each company that is coming and that you want to apply to:

1) Prepare a document that lists all the prominent products the company makes
2) For each product, list down some advantages and disadvantages
3) For each product, determine the business model

While you are preparing the document, you don't need to go into too much of depth, but preparing for questions like:

Name a MIcrosoft Product that you have used and don't like. What can you change about the product?

requires you to name atleast a few of the products and their disadvantages. For the question asked above, you might be wondering, "*What if I don't use any Microsoft Product?*"

I was in a similar situation during preparation, and I realized that the only ways out were to look at Youtube videos of how Microsoft products work and read online reviews about the various products. And of course I used Microsoft Office. And most probably you do to. And maybe you had used Internet Explorer in the past or were using a Windows PC before you migrated to a Mac. In that case, get to know the advantages and disadvantages of Microsoft Office (Word, Powerpoint, Excel), Internet Explorer, and Windows OS as compared to its competitors.

For Microsoft specifically, you can refer to the list of products below while preparing for their PM role:

1) Windows
2) Microsoft Office (Word, Powerpoint, Excel) + Office 365
3) SQL Server
4) SharePoint
5) Azure
6) Edge Browser

7) Skype
8) Bing
9) XBox
10) Hotmail
11) Cortana
12) LinkedIn
13) Hololens
14) Surface Pro
15) Microsoft Mobiles
16) Teams/MSN Messenger/Microsoft Forms etc.

If you are targeting Amazon, it's highly recommended that you learn their flywheel and their 14 leadership principles. The 14 leadership principles are:

1) Customer Obsession
2) Ownership
3) Invent and Simplify
4) Are Right, A Lot
5) Learn and Be Curious
6) Hire and Develop the Best
7) Insist on the Highest Standards
8) Think Big
9) Bias for Action
10) Frugality
11) Earn Trust
12) Dive Deep
13) Have Backbone; Disagree and Commit
14) Deliver Results

The flywheel is shown in the diagram below:



More about Amazon and its principles can be found at the following link:
https://www.amazon.jobs/en/principles

For other companies, it is advisable to go to their respective websites and find out the products they sell and how they make money.

Chapter 12

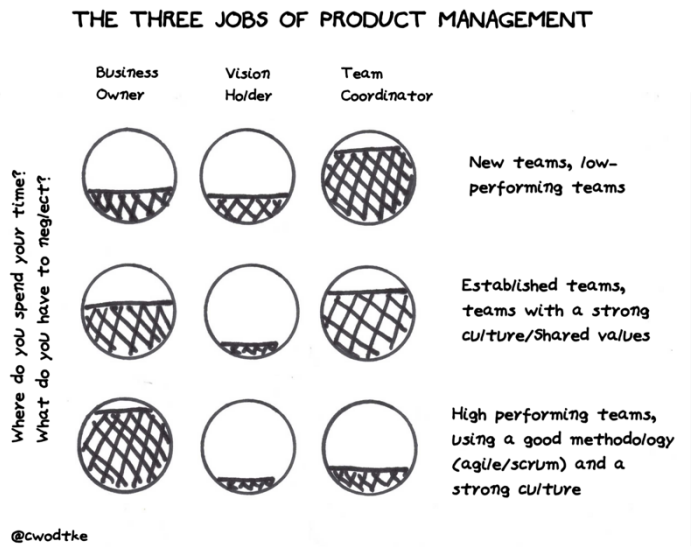
Conclusion

Wow! After what seems like ages, we have reached the conclusion of this book. I hope that through the gyan given here, you have gotten a more clear picture of what goes on inside the mind of a product manager.



Image Source: https://cdn-images-1.medium.com/max/1200/1*H95IU8954Gr0OsRoJzEsTw.jpeg

Product management is an interesting role and depending on the company you are going to work in, the role will change scope.



Source: https://cdn-images-1.medium.com/max/1600/1*0Uc89VQDDtzCu3a6EfLW3Q.png

And by any chance were you a developer or a coder before this? Didn't you hate those times when some guy just walked up to you and ordered you to design software that didn't make sense? Guess what. Once you are a PM, you will have a chance to do that. You can walk up to a random engineer and tell them to do technically unfeasible things.



Source: https://rahulabhyankar.files.wordpress.com/2017/04/0e0d21c.jpg?w=700

And so, with all that gyan passing over, let's call it a day. Good luck with preparing, and hope the information passed here was useful. Signing out.

Appendix 1 - GD Day Cases

Many PM companies don't carry out GDs. Though some, like Microsoft and Sprinklr, do. The GDs of both the companies operate very differently. Sprinklr carries out a typical GD. The Sprinklr GD topic for me was 'Design a unique method to solve Bangalore's traffic problem'.

Microsoft's GD is very different from the GD of other companies. Microsoft's GD is not strictly a GD. It is a new product development session in which you will be given a problem statement and you will have to come up with a new product and its specifications in a limited amount of time.

My Microsoft GD topic was: "Develop a travel solution for the non tech-savvy population of India". In one hour, we had to come up with a target market, maybe some numbers, product-specs, revenue-model, and a 1-minute speech explaining everything we thought off in that one hour.

My approach to the question was as follows:

Thoroughly understand the problem: I broke down the main problem into several questions that I could ask the invigilator. What does travel mean? Does it mean travel to visit family? Or religious travel? Or travel for work? What does non tech-savvy mean? Does it mean people who don't have phones? Or does it mean people who have phones but can't use them? Kids? Senior Citizens?

Identify Users & Customers: The invigilator answered by questions that travel could be for any purpose, as long as it was being done by the non-tech savvy population of India. I specified the market to be the rural population of India, adult males and females between the age of 21-65, who are travelling for religious + work reasons. Doing a simple guesstimate to put a concrete number to the target market, we can carry out the following steps:

1) The rural population of India is around 70%
2) The adult population of India is about 50%
3) I assumed smart-phone penetration in India to be around 25%

With India's population being 1.2B at the time of this book being written, the number comes out to be 1.2B x 70% x 50% x 25%, which is about 100M target users.

Understand the user journey: How do people use travel apps as of now? They search for it on the Play store (We are focusing on Android users here), and then download it. Then they register on the app using their phone numbers, and then carry out the operations such as searching for destination, choosing the best option, planning the trip, and making the payment. I didn't have

too much time to make fancy diagrams for this part. I just had enough time to get the overall journey right.

Identify pain points/gaps in the existing product: For non-tech savvy people there are many gaps in using the current apps. What is the crux of being non tech-savvy? It means that a person is uncomfortable in using regular technology. Thus, too much clicking and having too choose from too many options on screen might be confusing to them. Also, they might not have credit/debit cards or even mobile banking, so current payment methods might be an issue. Finally, the entire interface is in English, and being non-tech savvy there is a high possibility they don't know English well. The UI will then be an issue as well.

Prioritize pain-points: The primary problem to solve is that of accessibility and usage of the application. It has to be easy and intuitive to use, and requiring minimal interface and inputs from the user side. Also, the UI has to be understandable by any user. If these non-tech savvy people are located in remote areas, their internet speeds might not be good, which means that the app has to be light in weight and usage.

Generate solutions: What is the best solution to minimal user interface and easy access by non-tech savvy people? Voice control. And since the GD is being carried out by Microsoft, using Cortana would be the best way forward. Along with voice control, the UI will be minimal, with as few buttons as possible and all well spread out and decluttered. This will also reduce the space the app takes and maybe make the app require less internet data. Also, on phones with a low amount of RAM, running light applications is easier and less-prone to crashing.

To solve the other problems, the app will be available in local languages and not only in English. While searching for routes from point A to point B, the cheapest route will be favoured over the quickest route, keeping in mind the target market. Also, trains will be favoured over planes. Maybe planes won't even feature in the search options.

Pricing Model: How are we going to price the app? Similar to how current travel apps work. A commission will be charged during every transaction the user makes. In case the user can only pay by cash, we will identify local shops in each village that will take cash from the user and transfer the amount to us. This will solve the user's payment problem.

Go-to-market: The app will be advertised on the local radio in the local language. Also, small A4/A3 sized posters will be pasted in the local shops and community centers talking about the app. Small groups of advocates can be sent out to talk about the app and show the users its benefits.

Success Metrics: How will we measure success of the app? Since it will be a new app, we can use

metrics like 1) Number of unique downloads per week 2) Number of registered users per week 3) Number of bookings made per week 4) Total commission earned per week. All these metrics will be relative over multiple time periods, showing us the performance of the app over time.

Of course, better solutions are possible to this problem, but given the time constraints, it is better to have a complete solution ready and to aim for the best possible solution

Appendix 2 – One Day in the Life of a PM Intern

Company: Sprinklr

Intern: Pradeep Kumar VB

Sprinklr provides a gamut of digital solutions built on a unified platform which primarily focuses on several front office (customer-facing) platforms. The digital solutions are bundled under five clouds namely – modern marketing, modern advertising, modern research, modern care, and modern engagement. They are highly successful in the market due to two main reasons – the unified platform written using a single code base and their powerful AI solutions/features in the various clouds.

All of the interns were offered two projects. For the first project, a particular industry was assigned, and the various business drivers of the industry had to be analysed. Though it initially sounded like a consulting project, we got opportunities to interact with the various teams of Sprinklr to understand their solutions and how the industry's pain points could be solved. By understanding the emerging industry trends and the technology provided by Sprinklr, various business cases which can be powered by Sprinklr were also formulated.

The second project was much more closely related to that of a product manager's responsibilities at Sprinklr. I worked on creating an ROI framework for a particular feature in Sprinklr's Research cloud. I performed a competitive analysis and recommended features which could improve the ROI of the feature. I also closely worked with a Senior Product Manager to create a presentation template which would be used by the sales team for pitching to our clients.

The other projects given were –

1) Implement a new feature by creating a product concept, persona-based use cases, wireframes and mocks for usability/UX, GTM strategy. Additionally, sales pitch decks and marketing flyers were also designed.
2) Work with the AI and Advertising cloud team to collect and analyse data from Facebook advertising campaigns and layout algorithms to improve the campaigns' performance.
3) Providing recommendations to increase the usability of the platform after interacting with various clients.

Interns are expected to drive their projects on their own and there wouldn't be any training program or a mentor who can provide suggestions. Interns are given full freedom throughout the project,

but the suggestions provided by interns are generally implemented into their products (if you are able to convince them, of course).

Due to the flat hierarchy, interns get to interact and present their work to top executives like CTO and VP. With a start-up culture embedded in the organization, the employees (yes, that including the interns) are provided with flexible office hours, fun parties and indoor games.

Being leaders in the latest customer-facing AI technologies, Sprinklr offers a steep learning curve. PMs get complete ownership of projects in a span of 6 months. It is definitely an exciting place for a Product Management Role.

A basic grasp and passion for their technology along with strong collaboration and communication skills would definitely help to get through their internship selection process. Please reach out to me (pradeep.kumarvb18@iimb.ac.in), if any assistance is required.

Company: Amazon

Intern: Abhishek Mahato

To understand what happens in a day of an Amazon employee you need to first understand the company culture and the way of running things. Amazon is highly customer focused and this particular leader principle (out of the 14) is the one which dictate all the major business and other important decisions in the company. The other important aspect is that the company has a no presentation policy and all decisions, internal pitches and meeting are based on documents.

My daily schedule was based on the following caveats:

1. My project was to conceptualize a new product and the deliverable was the document which would be cover the business aspect of the product.
2. My assignment was an isolated project with no overlaps with the day to day working of the team to which I was assigned.
3. My manager did not give me work on the side while I was working.

I had divided the total work into weekly tasks and used to follow the timeline. During the initial half of the internship was to under how the business works from the internal and external stakeholders to understand how the product is of use to the customers. Thus I would have meeting, lined up in the second half of the day when the employees were usually free to give me time. And the first half was spent in building the document. I started working on my document quite early since I did not have side assignments and went through multiple iterations.

The first half of the internship was spent in developing the first draft of idea for the product. The 2nd half was spend in incorporating the feedback which had been given. The schedule of any given day remained the same as the first half but now the frequency of the meetings reduced with more time spent working on incorporating different perspectives and improving the document. There was more solo working involved during the latter half on my internship.

Lastly, as an intern I wasn't involved in the development and implementation of the product which is an integral part of the responsibilities of a permanent employee who is a product manager.

Company: Microsoft

Intern: Abheek Das

Being a PM intern at Microsoft is one of my best experiences working in a corporate setting. I interned at the Bangalore office of Microsoft IDC during my summers. It was an amazing learning as well as a lot of fun.

The topic of my summer internship project was to analyse Microsoft's internal ML platform, benchmark it against other similar internal and external platforms and come up with a set of recommendations for the betterment of its platform, make it more competitive in the market and make the internal data scientists much more agile in deployment of ML in Microsoft products.

A Typical Day

Though there are no set workhours as such mandated by Microsoft, mine usually started at around 8:30 in the morning and ended at about 5:00 in the evening. I usually had around one to two meetings scheduled with customers, data scientists, developers or senior PMs to understand the current solution that was present and identify some of the pain points and current caveats that needed to be solved. For me, a major portion of the learning & project progress happened through these interactions. The rest of the time was usually spent in reading internal documentation from various sources as well as doing a market survey by reading up competitor's developers guide, research papers, tech blogs & attending brown bags etc.

Other than that, I also had hands-on experience of their internal platform and deployed a few ML models end-to-end as experiments to establish proof of concept for my recommendations. This required working closely with the dev team. Finally, the day usually ended with me touching base with my manager by updating him on the progress and getting some feedback regarding any course-correction or discussing the next steps in the project.

Other Interesting Opportunities:

I was fortunate enough to participate in the Hackathon and the Ideathon with my fellow teammates. These events/competitions are conducted frequently at Microsoft and you can learn about a lot of interesting ideas from fellow employees and gave a flavour of the life at Microsoft.

Also, for PM interns there were Leadership Sessions organized by HR, where we got an opportunity to have 1:1 interaction with senior leaders at Microsoft. They gave a lot of valuable insights into the roles and responsibilities of a PM, explain the company's long-term vison in the tech. industry, as well as good career advice. These sessions are a must attend for anybody looking for a long-term career in PM.

The Fun Stuff & Perks

Though there were lots to be done in the summer internship but rest assured that there is ample time and opportunities to relax and let your hair down. One can de-stress by playing Xbox in 4K screens or simply over a friendly game of Foosball or Pool. If you are the outdoorsy types, there's football, volleyball or basketball which are always open. I liked to usually end my day with a workout at the Cult Fit Gym which was open to all employees and interns alike.

I had one big team outing at an exotic location, where we had multiple team bonding exercises and team lunches where we could interact with other PMs in the org. at all level and get insights into the job & life at Microsoft.

Overall the whole summer internship was pretty chilled out and I got ample help from my teammates and manager to complete the project successfully and to the best of my potential. The general work culture is pretty open, and everybody is ready to help out & share their knowledge which made the summer internship an amazing experience.

Appendix 3 – Actual Interviews Experiences

Name: Nikita Nallamothu

Company: Amazon (MLDP), Summer 2018

1. Describe the steps in the process i.e. no of interview rounds, GD, pre-process requirements (LOI, case submission etc).
   Once Amazon visits the campus and opens up the roles they are going to offer at the summers, students can apply for maximum 3 roles. Amazon came for around 8 roles last year. Amazon does not conduct GD. I had 2 rounds of interviews. It was a mix of technical and HR.

2. Does work experience matter?
   Yes, Amazon has a criteria for work experience while application. Only the Category Manager role does not require work ex.

3. What kind of questions were asked? What is the general area of focus during the interviews?
   In the first round of interview I was about my work experience. They focused on situations where I showed my leadership skills and resolved business issues. I was also asked about how my learnings from my work experience can be implemented at Amazon. The second interview was a product design question asked in product management interviews. I was asked to design a system for the government for waste management in a metropolitan city.

4. What kind of HR questions were asked?
   The first interview was a kind of HR round. I was asked only about work experience. The leadership initiatives I had taken up, and difficult situation that I overcame and how.

5. How do you recommend a pgp1 should structure his preparation? Resources/ tools that can help aid in preparation for interview?
   Be up-to-date with all the recent happenings in the tech world particularly ecommerce sector. Prepare guestimates, it will be useful for any interview. For PM roles definitely read the book "Cracking the PM Interview" by McDowell and Bavaro. It is important to have interest in technology and it should come out through your interview

6. How was the internship experience, work culture and PPO policy?
   There is a PPO policy.
   Internship experience: My work was related to Business Development. I had a good support from my Mentor and team. Working hours are flexible and work from home is possible depending on your manager and mentor. The Interns are expected to be accountable for their project. We were given the project objectives and deliverables at the very beginning so that we

could set up weekly milestones. Candidate has to be good at written communication as everything including new business plans/product or feature launches will be presented only in the form of a 6-page narrative and PPTs are not used at Amazon.

7. What was your role in the company?
   My role was Marketplace Leadership development program and my work was related to business development

8. Any Feedback/comments for PGP1s
   For PM role pick up products and think about how you can improve them. Think from customer's point of view. Be aware of the recent happenings in e commerce. My interview was very similar to PM interview. Be thorough with your resume. Be ready to explain each and every point. Prepare about your past work ex properly.

9. What were your key learnings from this internship?
   Knowledge about the business model and functioning of ecommerce. I got to understand the important metrics that are emphasized upon in the company.

10. How do you recommend a pgp1 should structure his preparation? Resources/ tools that can help aid in preparation for the internship?
    Answered above

Name: Sucheth P

Company: Amazon (MLDP), Summer 2018

1. **Describe the steps in the process i.e. no of interview rounds, GD, pre-process requirements (LOI, case submission etc).**
   Work experience required: Yes
   Number of interview rounds: 2
   GD: No

2. **Does work experience matter?**
   Yes. Applicants should have min 21 months' work ex.

3. **What kind of questions were asked? What is the general area of focus during the interviews?**
   First Round:
   The interview was a short one with only one interviewer. He gave me a case question stating, "The sales of Trousers are declining continuously, find out the reason". After this he asked me if there is any new feature that I can suggest for Amazon's website.

   Second Round:
   The interview started with my past work responsibilities. As I have worked for an automobile company, they told me to design a mobile app for showcasing important features of car. They told me focus on user interface and the important features (like car's fuel level, engine oil level etc). The discussion continued to ways of earning revenues from the applications and key performance indicators. Also, some questions revolved around how the data retrieved from each user be useful for business purposes. The discussion went for around 30 minutes.

4. **What kind of HR questions were asked?**
   Any ethical dilemma faced in work environment? How did you handle any conflicts you faced with your supervisor?

5. **How do you recommend a pgp1 should structure his preparation? Resources/ tools that can help aid in preparation for interview?**
   While answering always follow a structured way. The deeper you drill down any questions, the better it is. I had a very healthy discussion in the case interview. The interviewers were helping a lot in between. Go through Amazon's 14 principles.

6. **How was the internship experience, work culture and PPO policy?**
   I was posted in Marketplace business. I was an MLDP intern. Personally, I had a very nice project to work on. I worked on Amazon Fashion division. The project aimed at finding reasons for decline in one of the category's sales.

Work culture is very dynamic. I used to work from 10 am to 8 pm. Nothing is spoon fed. You are expected to work starting from day 1. Mentor allocated will be very busy, hence you should go and ask them your doubt, else you would find difficulty. Amazon doesn't have ppts, instead at the end of your project you need to submit a 6 page report. The supervisors for evaluation will consists of very senior managers like – VP of the division.

PPO policy is there. Conversion largely depends on your performance and how well you align yourself with the leadership principles of Amazon.

7. What was your role in the company?

   I was a MDLP (Market place leadership program) intern. MLDP is kind of an internal consultant. If you get a PPO, then you will be revolved around various divisions of Amazon, assisting on their projects. For internship I was a Category BDM, - where my main motive was to identify the gaps in the decreasing sales of a category and recommend solutions to ramp up the sales. The project involved data analysis, digital marketing and business strategy concepts.

8. Any Feedback/comments for PGP1s:

   Interview : Be calm and confident in the interview. Try questioning the interviewer to gain more information during case solving, don't just give answers based on your assumptions.

   Internship: Deliver results which are backed up by data. Don't globe. Do well in mid term review.

9. What were your key learnings from this internship?

   E commerce business models, Consumer purchasing behaviour, supply chain of e commerce.

10. How do you recommend a pgp1 should structure his preparation? Resources/ tools that can help aid in preparation for the internship?

    Guestimates are must. Learn about e commerce models. FDI rules of e commerce. Learn excel if get selected.

Name: Swapnil Katiyar

Company: HUL- IT, Summer 2018

1. Describe the steps in the process i.e. no of interview rounds, GD, pre-process requirements (LOI, case submission etc.)
   Initial interest form, GD shortlist, GD on GD day, Day 1- 2 Rounds of Interview.

   GD Case based: a caselet (1.5 pages) for rural ERP system and ways to improve it. 5 min to read + 20 to discuss + 5 to conclude. focused mainly on the ideas when aforesaid constraints given. Attending the preplacement talk helped a lot as they mentioned key focus areas in that which helped steer the discussion.

   Interview R1 (1hr+): It was quite extensive starting from tell us something about yourself to analyzing almost all resume points from my work ex. To me it seemed they were trying to gauge my interest and understanding in IT solutions as a business instead of developer. "Why HUL-IT and not any other Prodman company? (They knew that I just had an interview with Microsoft, so grilled on that)".

   Round 2 (25min): I was asked to wait for a while before they announce results but instead they called me in for another round of interview. This had only one question- "Why HUL-IT and not any other Prodman company?"

2. Does work experience matter?
   Yes, but not mentioned as a necessity. Its mainly regarding the understanding of entire machinery which comes with work-ex.

3. What kind of questions were asked? What is the general area of focus during the interviews?
   Refer q1. Majorly focused on application in real life and overall understanding of IT solutions.

4. What kind of HR questions were asked?
   Why IT role in FMCG and not Prodman in core product based company?

5. How do you recommend a pgp1 should structure his preparation? Resources/ tools that can help aid in preparation for interview?

6. How was the internship experience, work culture and PPO policy.
   IT role is global function and it is for parent company Unilever and not HUL so my experience is bound to be a bit different from say marketing interns in HUL. My location was Bangalore, which is IT head office for Unilever in India.

   Most well-structured Internship in terms of clarity and support from the company.

Work culture is a bit team dependent but is overall fine and seems more like a services company (not in a bad way but because of a lot of consultants and external vendors). PPI is given first which is result of extensive feedback from the mentor and his/her line director along with feedback from the mentor assigned (VP level person). Who all got PPIs is announced on the last day at Mumbai along with other Interns and interviews are held then and there. Interview results are announced after a couple of weeks.

7. What was your role in the company?

   I was the champion for a cognitive automation initiative involving chat bots for Fusion team supply chains (India, Nepal and Srilanka). The role closly resembled Product manager as I was handling inception to delivery but that might not be the end role I join.

   After PPO, one gets into a pool of resources and goes through a training of 1.5 years approx., called UFLP. After that interviews happen to assign the relevant and most suited role.

   I found the role closely resembling a GenMan Profile owing to lot of stakeholders interactions and vendor management as well.

8. Any Feedback/comments for PGP1s:

   Target companies well based on your forte and read up about them and their latest Initiative. Attending PPts help a lot. Take notes, if not you, your friend might need it.

   Have a good understanding of your workex and its business implications. And form Prodman GD groups separately as they are very different from normal GDs.

9. How do you recommend a pgp1 should structure his preparation? Resources/ tools that can help aid in preparation for the internship?

   Mostly covered above.

Name: Kedar Mahamuni

Company: VISA, Summer 2018

1. Describe the steps in the process i.e. no of interview rounds, GD, pre-process requirements (LOI, case submission etc).
   There was no SOP or case submission.
   There was one GD round. The GD was quite normal. In my GD round, they gave topic on demonetization.
   There were two rounds of personal interviews, both were technical.

2. Does work experience matter?
   Yes. The minimum work-ex requirement for application was 42 months.

3. What kind of questions were asked? What is the general area of focus during the interviews?
   In both the rounds, lots of questions on work experience were asked. As I had 4 years of software engineering experience, they were more interested in knowing what type of challenges I faced, and how did I provide technology-based solutions.
   There were few product design questions as well. Also, they expected end to end technological solutions in detail. Some questions on security algorithms were also asked.
   Few questions on latest happenings in technology domain were asked

4. What kind of HR questions were asked?
   No HR questions were asked to me.
5. How do you recommend a pgp1 should structure his preparation? Resources/ tools that can help aid in preparation for interview?
   The preparation should involve practice of product design questions.
   As most of the questions are from previous job, so proper understanding about things written in CV is essential.
   Subscribe to newsletters provided by TechCrunch, Wired, Inc42, McKinsey Insights and the like where you will know the latest technologies and their applications in various firms around the world.

6. How was the internship experience, work culture and PPO policy?
   Internship experience was really good. Visa truly believes in data-driven decision making. Culture was also really good. Everyone was supportive and ready to help you. You will get to interact with a number of internal teams to get your data or analysis done. Though there were few uncertainties about the scope of the project, later they helped us to narrow down the scope.
   The usual working hours used to be from 9 to 5. Never had to work on weekends as well. They had also provided a good quality accommodation for entire 2 months.

PPO Policy: Yes (Yet to be released for last year summers)

7. What was your role in the company?

   My role was Technical Product Manager. This role demands strong technical understanding along with product management skills.

8. Any Feedback/comments for PGP1s:

   Not just about VISA, but for other tech companies also, practice guesstimates and product design questions. Keep calm during your interviews. Most of the product management interviews tests your problem-solving skills. Also, I would recommend to go through DS-1 and DS-2 concepts.

9. What were your key learnings from this internship?

   I got to learn many things during internship like role and responsibilities of product manager. Product management is marketing and analytics intensive role. As VISA works in payments domain, I got to learn a lot about payments industry and innovations in this industry as well.

Name: Nukala Vijaya Lakshmi

Company: Microsoft IDC, Summer 2018

1. Describe the steps in the process i.e. no of interview rounds, GD, pre-process requirements (LOI, case submission etc).
   1 GD
   1 Interview

2. Does work experience matter?
   No

3. What kind of questions were asked? What is the general area of focus during the interviews?
   GROUP DISCUSSION TOPIC

   How LinkedIn can be used to in integration with other Microsoft applications to leverage it?
   Step 1: The main consumers I am keeping in mind while doing this integration is students and corporate professional who are supposed to do power point presentations and word documents frequently as part of their presentations to the companies or for learning. So here we use Microsoft word document + Linked IN and Power point presentation+ Linked in.

   Step 2: Idea is when a student is working in Word document about a project then according to the key words he types in the document there is a AI search and the results related to the relevant data like power points, documents, and links etc., related to the topic were displayed in a tab, this information is derived from the scholarly data base present in LinkedIn. So that the data base can be used in the LinkedIn to leverage. And Microsoft has a feature called "INSIGHTS" which will summarize the transactions of the data related to bank accounts etc. So here if the professional or student doesn't have time to skim through all the references that we provide in the references then insight which have a summarized version of the content can be used to refer and get relevant information from that. This insights tab will be provided in the word document

   Step 3: Short comings of this process
   There is a lot of information in database and user will be overwhelmed with the information provided and gets confused as what to take and up to which extent he should take from the database.

Step 4: How to eliminate this
This can be eliminated by creating options like Student, corporate, Scholarly research etc., so that the relevant information will be provided based on the option we click for. And the quality of the articles will be measured based on the feedback like shares, likes, and comments and ratings given to those articles and they are categorized in to those sections according to its score and the profile of the author of the source article

Step 5: Future updates
If necessary, any videos or conferences related to it can be sent as notifications to the users concerned. So that users can go to the relevant events and gain knowledge from them

INTERVIEW EXPERIENCE
Topic: How will you design a chess game for players
Step 1: First my consumers will be
1) Beginners who have interest in chess and want to learn
2) Players who have a minimum knowledge
3) Tournament players and above

Step 2: In case of allocation of player, we will allocate player based on path dependency for example if a player has won certain games with each game he will be getting some points and based on the points all the players are categorized in to different levels and opposite player will be allocated to a player based on the scores with in those levels. The levels were categorized based on the number of wins for the game, any ratings like Chess federation ratings and some key steps which are used during games which are used by players like Carlson, Anand etc., If the player lost for more than 2 games then the player to him will be allocated a level below to him so that his chances of winning will be high, and he will get back on track with confidence. Here after the game was completed recommended suggestions will be displayed as how Carlson would have played if he was in the players position. How better the player can do the moves and where he can do and analysis for his failure and success.

Step 3: Disadvantages or short comes
Here a player will be allocated other player based on his scores, but back end work will be high during allocation of players. There will be lot many players and the algorithms processing will become difficult to get the right player as pair.

Step 4: To eliminate this
We should divide this based on geographic locations and initially the player should be allocated with players of same geography then with the performance he should be allocated with other geographic players if he is topping in his own geographic regions.

Step 5: Future updates
In future we can add features like notification to the players regarding events in and around the world about the games and information regarding any contest that are held in chess game and videos links of tournament players as recommendations.
Interviewer: what are the key metrics you will use to evaluate the performance of this project
Me:
1) I am assuming this as an app. So, I will consider the number of downloads, number of shares they do this app

2) I will consider the time spent by a player typically using this app

3) I will consider the feedback ad recommendations made about the game

4) I will also consider how effectively people are using to utilize the notifications of events updates we are providing them

Note: I have resume point in Inter University level chess championship. The interviewer saw the resume and asked to design the game

4. What kind of HR questions were asked?
   None

5. How do you recommend a pgp1 should structure his preparation? Resources/ tools that can help aid in preparation for interview?
   Focus on latest technological developments and keep abreast with changes in products.